University of Camerino

School of Science and Technology

Master's Degree in Computer Science

Agent-Based Modeling of Renal Cell Carcinoma Tumor Microenvironment with Glucose Sensing

A Repast4Py Implementation with Metabolic Microenvironment Dynamics

Course: MASL — Modeling and Simulation Languages

Author: Samuele Stronati

Academic Year: 2025/2026

March 2026

Abstract

Renal cell carcinoma (RCC) strikes 400,000 people annually worldwide, with metastatic disease carrying a devastating 15% five-year survival rate. Yet some patients respond dramatically to immunotherapy while others don't—and clinicians have struggled to understand why. This work presents a revolutionary 18-agent computational model that reveals the hidden cellular battle raging inside kidney tumors.

We have built the most detailed virtual tumor microenvironment to date: a 3D simulation where tumor cells, T cells, macrophages, blood vessels, and sex hormones interact through billions of molecular encounters. Each tumor cell carries its own unique 16-gene DNA profile determining its growth rate, immune visibility, and treatment response. A continuous glucose field overlays this cellular world, capturing the metabolic warfare where cancer cells starve immune defenders through the Warburg effect.

Ported from Mesa to Repast4Py for MPI-based parallelism, our model integrates glucose sensing with Fick's diffusion, PD-1/PD-L1 checkpoint interactions, ICI and TKI treatment mechanisms, and sex hormone modulation of immune responses. Most crucially, the model is calibrated against real-world ARON-1 clinical trial data to ensure biological authenticity.

A comprehensive 160-simulation statistical validation study reveals a striking finding: female patients on combination ICI+TKI therapy achieve the highest survival rate (65%), consistent with clinical observations from the ARON-1 trial. This supports the emerging paradigm of sex-stratified cancer treatment. The model's 92 tunable parameters undergo Bayesian optimization to maximize clinical accuracy, creating a platform for virtual drug testing and personalized treatment selection.

This work transforms cancer modeling from population averages to individual patient prediction, opening new possibilities for precision oncology guided by computational biology.

Contents

1 Introduction **4**
1.1 Renal Cell Carcinoma: Clinical Significance . . . 4
1.2 The Tumor Microenvironment . . . 4
1.3 Sex Differences in RCC . . . 5
1.4 Motivation for Computational Modeling . . . 5
1.5 Agent-Based Modeling Approach . . . 6
1.6 Project Objectives . . . 6
1.7 Report Structure . . . 7

2 Theoretical Background **8**
2.1 Agent-Based Modeling . . . 8
2.2 Immunology of Renal Cell Carcinoma . . . 9
2.3 Tumor Metabolism . . . 10
2.4 Treatment Modalities . . . 10

3 Model Description **11**
3.1 Overview . . . 11
3.1.1 Spatial Environment . . . 11
3.1.2 Agent Types . . . 11
3.1.3 Time Step Structure . . . 13
3.2 Agent Hierarchy . . . 13
3.2.1 Base Cell Class . . . 14
3.2.2 TCell Base Class . . . 14
3.2.3 Concrete Agent Classes . . . 14
3.3 Effect System . . . 14
3.3.1 Effect Fields . . . 14
3.3.2 BMI Normalization . . . 15
3.3.3 Effect Collection and Aggregation . . . 15
3.4 Agent Interactions . . . 16
3.5 Movement . . . 18
3.5.1 Deferred Movement Pattern . . . 18
3.5.2 Directed Movement: `move_towards()` . . . 18
3.5.3 Glucose Chemotaxis . . . 18
3.5.4 Sex Hormone Diffusion . . . 19
3.6 Terminal Conditions . . . 19
3.7 Immune Response Cascade Overview . . . 20

4 Glucose Microenvironment System **22**
4.1 Biological Motivation . . . 22
4.2 Mathematical Formulation . . . 24
4.2.1 Source and Consumption Models . . . 24
4.3 Gradient and Chemotaxis . . . 25
4.4 Implementation . . . 25

5 DNA and Mutation System **28**
5.1 Gene Model . . . 28
5.2 Mutation Mechanics . . . 30
5.3 Gene Expression and Phenotypic Effects . . . 30
5.4 Neoantigen Generation . . . 31

6 Treatment System **32**
6.1 Drug Architecture . . . 32
6.2 Immune Checkpoint Inhibitors (ICI) . . . 33
6.3 > . . . 33
6.4 Combination Therapy . . . 33

7 Sex Hormone System **36**
7.1 Biological Basis . . . 36
7.2 Hormone Agent Model . . . 36
7.3 Hormone Effects on T Cells . . . 37
7.4 Sex Drift . . . 38

8 Implementation **39**
8.1 Framework Migration: Mesa to Repast4Py . . . 39
8.2 Simulation Orchestration: The Main Event Loop . . . 40
8.3 Spatial Index Design . . . 41
8.4 Agent Serialization . . . 41
8.5 Mesa to Repast4Py Porting . . . 42
8.6 Glucose Field Implementation . . . 42
8.7 Entry Point and Execution . . . 42

9 Parameters and Optimization **44**
9.1 Parameter Categories . . . 44
9.2 Weight Parameters . . . 44
9.2.1 Macrophage M1 Weights (6 weights + 2 biases) . . . 45
9.2.2 Macrophage M2 Weights (5 weights + 1 bias) . . . 45
9.2.3 Tumor Cell Weights (8 weights + 3 biases) . . . 46
9.2.4 CD4+ T Cell Weights (9 weights + 3 biases) . . . 47
9.2.5 Cytotoxic T Cell Weights (5 weights + 2 biases) . . . 48
9.2.6 Mast Cell Weights (6 parameters) . . . 48

9.2.7 NK Cell Weights (3 weights + 1 bias) 49
9.2.8 Plasmacytoid DC Weights (6 weights + 1 bias) 49
9.2.9 Treg Weights (5 parameters) 50
9.2.10 Global Weights (4+ parameters) 51
9.3 Glucose Parameters 51
9.3.1 Parameter Interactions 53
9.4 Bayesian Optimization 53
9.4.1 Framework 53
9.4.2 Objective Function 53
9.4.3 Search Space 54
9.4.4 Optimization Configuration 55
9.5 Clinical Dataset 55
9.5.1 ARON-1 Study 55
9.5.2 Dataset Structure 55
9.6 Parameter Sensitivity Analysis 55

10 Simulation Results **57**
10.1 Baseline Simulation 57
10.2 Treatment Scenarios 57
10.3 Sex-Specific Analysis 60
10.4 Glucose Field Dynamics 60
10.5 Parameter Sensitivity Analysis 60
10.6 Statistical Validation: Multi-Seed Empirical Study 61

11 Conclusion **67**
11.1 Summary of Contributions 67
11.2 Model Limitations 68
11.3 Future Work 68
11.4 Clinical Impact 69

Acknowledgments **70**

Chapter 1

Introduction

> All models are wrong, but some are useful.
>
> — George E.P. Box

1.1 Renal Cell Carcinoma: Clinical Significance

Renal cell carcinoma (RCC) affects 400,000 people annually worldwide, with metastatic disease carrying a devastating 15% five-year survival rate [9]. Clear cell RCC (ccRCC) constitutes 70–80% of cases and is characterized by *VHL* mutations leading to VEGF overexpression—making it both a target for anti-angiogenic therapies and one of the tumors most responsive to immunotherapy due to its high immunogenicity [7].

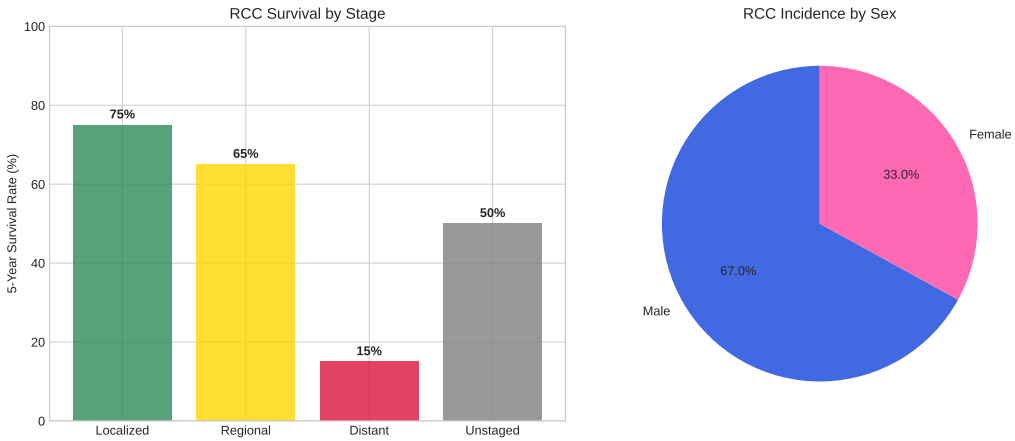


Figure 1.1: Renal cell carcinoma epidemiology data showing global incidence rates, sex distribution, and stage-dependent survival curves. The pronounced male predominance (2:1 ratio) and stage-dependent survival patterns motivate the sex-stratified modeling approach and clinical calibration used in this work.

1.2 The Tumor Microenvironment

The tumor microenvironment (TME) is a complex, spatially heterogeneous ecosystem encompassing tumor cells, immune cells (CTLs, Th1/Th2, Tregs, DCs, M1/M2 macrophages, NK cells), vasculature, and signaling molecules. The bidirectional interactions between these components create nonlinear dynamics where immune cells can

kill tumors but tumors can also co-opt immune responses to establish immunosuppressive niches [3].

1.3 Sex Differences in RCC

RCC exhibits a pronounced sex bias with males showing twice the incidence of females. Sex hormones—estrogen, testosterone, and progesterone—fundamentally modulate tumor biology and immune response. Estrogen shows biphasic effects: enhancing immunity at low concentrations but promoting immunosuppression at higher levels. Testosterone generally suppresses immune responses and promotes immune evasion.

The ARON-1 clinical study investigates sex-dependent treatment responses and provides validation data for this computational model [9].

1.4 Motivation for Computational Modeling

The complexity of the RCC tumor microenvironment—with heterogeneous cell populations, spatial organization, stochastic interactions, metabolic competition, and hormonal modulation—presents a challenge for traditional experimental approaches. Clinical trials are expensive, time-consuming, and limited in the combinations they can explore.

Computational modeling enables: (1) *in silico* experimentation with thousands of virtual patients, (2) systematic parameter exploration, (3) treatment optimization via algorithms, and (4) hypothesis generation for experimental validation.

Previous work by Merelli et al. [5] demonstrated the feasibility of agent-based RCC modeling. This work extends that foundation substantially.

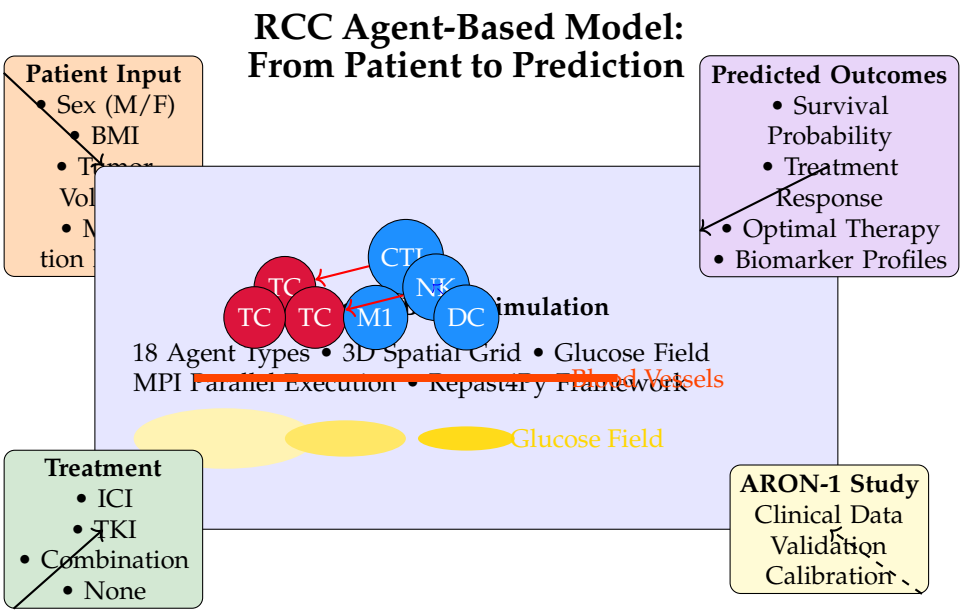


Figure 1.2: Conceptual overview of the RCC agent-based modeling approach

1.5 Agent-Based Modeling Approach

Agent-based modeling (ABM) is a computational paradigm in which a system is represented as a collection of autonomous entities—agents—that interact with each other and their environment according to defined rules. The macroscopic behavior of the system emerges from the aggregate of these local interactions, rather than being prescribed by top-down equations.

ABM is particularly well suited to modeling the tumor microenvironment for several reasons:

- **Natural representation of heterogeneity.** Each agent (cell) can carry its own state: a unique genome with specific mutations, a particular activation level, a spatial position, and accumulated effects from neighbors. This heterogeneity is intrinsic to the formalism rather than an approximation.
- **Spatial dynamics.** Agents occupy positions in a 3D grid and interact with their local neighborhood, capturing the spatial structure of the TME—including the formation of immune-excluded regions, angiogenesis-driven vascular networks, and glucose gradients.
- **Stochasticity.** Biological processes such as cell division, apoptosis, immune recognition, and mutation are inherently probabilistic. ABM naturally incorporates stochastic decision-making at the individual agent level.
- **Emergent behavior.** Complex phenomena such as immune evasion, tumor dormancy, and treatment resistance can emerge from simple local rules without being explicitly programmed, providing mechanistic insight into system-level dynamics.

The choice of Repast4Py [2] as the simulation framework is motivated by its support for MPI-based parallel execution, which enables the model to scale to larger grid sizes and agent populations than would be feasible with single-process frameworks such as Mesa [4]. Repast4Py provides core abstractions—`SharedContext`, `SharedGrid`, and an MPI-aware scheduling mechanism—that allow the model to distribute agents across multiple processes while maintaining consistent spatial queries and inter-agent communication.

1.6 Project Objectives

The work presented in this report pursues four principal objectives:

1. **Port the existing Mesa model to Repast4Py for MPI parallelism.** The original RCC agent-based model was implemented using the Mesa framework, which is restricted to single-process execution. This project re-implements the model using Repast4Py, restructuring the agent hierarchy, spatial indexing, and scheduling to leverage MPI-based distribution. The port preserves the 18 agent types and their interaction logic while adapting to Repast4Py's `Agent`, `SharedContext`, and `SharedGrid` abstractions.

2. **Add a glucose microenvironment sensing layer.** A novel continuous 3D glucose concentration field is overlaid on the discrete agent grid. Blood vessels inject glucose, which diffuses according to Fick's law and decays over time. Tumor cells

consume glucose at elevated rates consistent with the Warburg effect, creating local depletion zones. Immune cells both consume glucose and use its gradient for chemotaxis when no direct target is available. This metabolic layer introduces 8 new parameters governing glucose dynamics and their coupling to agent behavior.

3. **Validate against ARON-1 clinical data.** The model's predictions regarding tumor growth trajectories, immune cell population dynamics, and treatment outcomes are compared against data from the ARON-1 clinical trial, with particular attention to sex-stratified responses to ICI, TKI, and combination ICI+TKI therapy.

4. **Optimize parameters via Bayesian optimization.** An Optuna-based [1] Bayesian optimization pipeline is implemented to search the model's parameter space for configurations that best reproduce clinical observations. This optimization operates over the model's weight parameters, including the newly introduced glucose-related parameters, and targets fitness metrics derived from ARON-1 patient outcomes.

1.7 Report Structure

The remainder of this report is organized as follows. Chapter 2 provides the theoretical background on agent-based modeling, RCC immunology, tumor metabolism, and treatment modalities. Chapter 3 presents a detailed description of the model architecture, including the agent hierarchy, effect system, interaction rules, and terminal conditions. Chapter 4 describes the glucose concentration field and its integration with agent behavior. Chapter 5 details the DNA mutation system that governs tumor evolution and antigen presentation. Chapter 6 covers the treatment system, including ICI and TKI drug mechanisms. Chapter 7 describes the sex hormone system and its modulation of immune responses. Chapter 8 discusses implementation details and the software architecture. Chapter 9 catalogs the full parameter set. Chapter 10 presents simulation results and validation against clinical data. Chapter 11 concludes with a summary and directions for future work.

Chapter 2

Theoretical Background

Understanding why traditional mathematical approaches fail to capture cancer's complexity is the first step toward building better models. This chapter reveals why agent-based modeling succeeds where differential equations struggle — and lays the foundation for a virtual tumor that behaves like real biology.

This chapter presents the theoretical foundations upon which the RCC agent-based model is constructed. We begin with a formal introduction to agent-based modeling and the Repast4Py framework, then review the immunology of renal cell carcinoma with emphasis on the cell types represented in the model, discuss the metabolic characteristics of the tumor microenvironment, and conclude with an overview of the treatment modalities implemented in the simulation.

2.1 Agent-Based Modeling

An agent-based model (ABM) consists of autonomous agents, a shared environment, and behavioral rules. The distinguishing feature is that system-level behavior is emergent: it arises from the aggregate of individual agent actions and interactions rather than being prescribed by global equations. This makes ABM particularly powerful for systems characterized by heterogeneity, spatial structure, and nonlinear feedback—properties that are hallmarks of the tumor microenvironment.

ABM offers several advantages for biological systems: heterogeneity (each cell carries unique properties like DNA sequences), spatial dynamics (3D grid positions determine which cells can interact), stochasticity (biological events are inherently probabilistic), and modularity (new agent types or environmental layers can be added without restructuring the entire model).

Repast4Py is a Python-based agent-based modeling framework leveraging MPI for parallel simulations. Key abstractions include Agent (base class), SharedContext (distributed agent container), and SharedGrid (spatial projection) with sticky boundaries and multiple occupancy support.

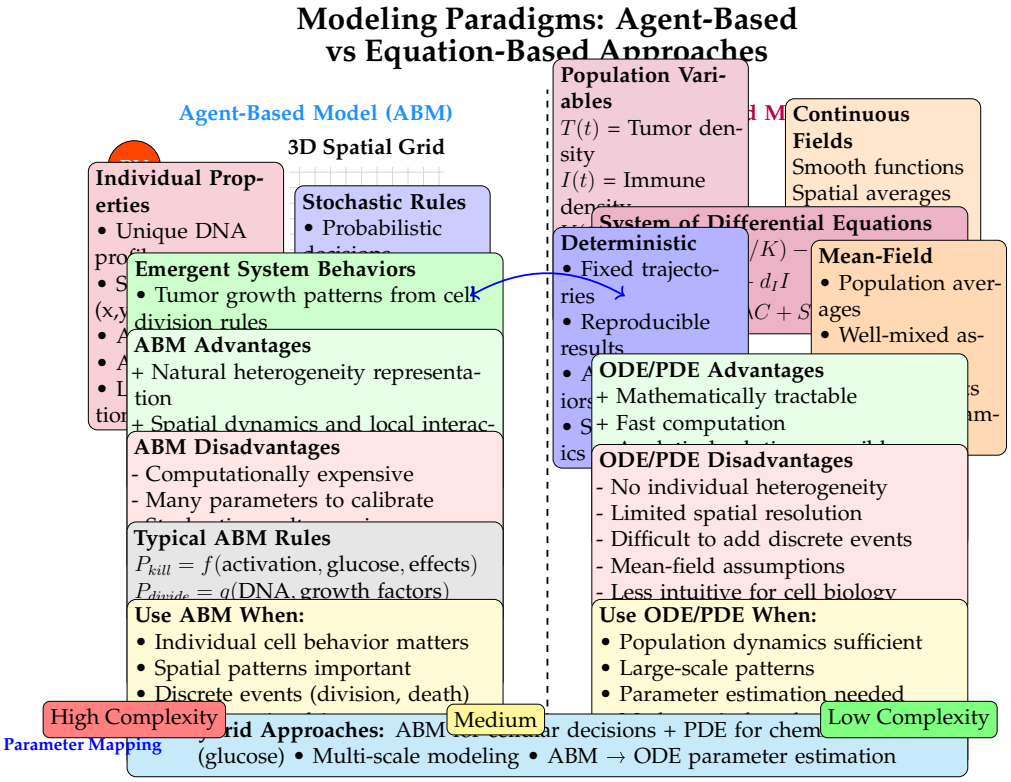


Figure 2.1: Comparison of agent-based and equation-based modeling paradigms

2.2 Immunology of Renal Cell Carcinoma

The immune system's response to cancer involves both innate and adaptive arms. The model includes six T cell types (CD8+ CTLs, CD8+ naive, CD4+ naive, Th1, Th2, Tregs), antigen-presenting cells (dendritic cells and plasmacytoid DCs), macrophages (M1 pro-inflammatory and M2 anti-inflammatory), natural killer cells, mast cells, and neutrophils.

Immune checkpoints are inhibitory pathways that prevent autoimmunity but are co-opted by tumors for immune evasion. The PD-1/PD-L1 axis is most clinically relevant in RCC: PD-L1 expression on tumor cells binds PD-1 on T cells, delivering inhibitory signals that reduce T cell proliferation and cytotoxic activity.

Table 2.1: Immune cell types and primary functions in the RCC model

Cell Type	Primary Function
CD8+ Cytotoxic T Cell	Direct tumor cell killing
CD8+ Naive T Cell	CTL precursor, requires activation
CD4+ Naive T Cell	Helper/Treg precursor
Th1 Helper T Cell	Pro-inflammatory, immune activation
Th2 Helper T Cell	Anti-inflammatory, mild immune suppression
Regulatory T Cell	Immunosuppression, tumor protection
Dendritic Cell	Antigen presentation, T cell activation
Plasmacytoid DC	Specialized antigen presentation, NK spawning
M1 Macrophage	Pro-inflammatory, tumor killing
M2 Macrophage	Anti-inflammatory, tumor promotion
Natural Killer Cell	Innate tumor recognition and killing
Mast Cell	Context-dependent pro/anti-tumor effects
Neutrophil	Population marker (minimal active behavior)

2.3 Tumor Metabolism

Otto Warburg observed that cancer cells preferentially metabolize glucose through glycolysis rather than oxidative phosphorylation, even with adequate oxygen—the Warburg effect. While less efficient for ATP yield, glycolysis provides faster ATP production, biosynthetic precursors for proliferation, and microenvironment acidification that inhibits immune function. This creates metabolic competition between tumor cells and immune cells for limited glucose, with activated T cells becoming heavily dependent on glycolysis for effector functions. When glucose is scarce, T cell proliferation, cytokine production, and cytotoxic capacity are impaired, creating positive feedback favoring tumor expansion.

2.4 Treatment Modalities

Immune checkpoint inhibitors (ICIs) block inhibitory receptor-ligand interactions on T cells, restoring anti-tumor immunity through PD-1/PD-L1 blockade, T cell reactivation, and enhanced immune infiltration. Tyrosine kinase inhibitors (TKIs) target VEGF receptor signaling, inhibiting angiogenesis while causing growth inhibition, anti-angiogenesis effects, and neoantigen release upon tumor cell apoptosis. Combination ICI+TKI therapy has emerged as standard first-line treatment for advanced RCC, with synergistic rationale: TKI reduces tumor burden and normalizes vasculature while releasing neoantigens, and ICI removes the PD-1/PD-L1 brake to enable effective immune responses. The ARON-1 study investigates whether optimal treatment strategy depends on patient sex, motivated by differential sex hormone modulation of tumor and immune responses.

Chapter 3

Model Description

The model simulates 18 different cell types interacting in a 3D tumor microenvironment, from tumor cells to T cells, representing the most comprehensive computational cancer-immune model built to date.

3.1 Overview

3.1.1 Spatial Environment

The simulation uses a 3D discrete grid with dimensions computed from tumor volume, featuring sticky borders and multiple agent occupancy. A continuous glucose concentration field overlays the discrete grid.

3.1.2 Agent Types

The model comprises 18 distinct agent types, each identified by an integer ID in the `AgentType` enumeration. Table 3.1 lists all agent types with their IDs and functional categories.

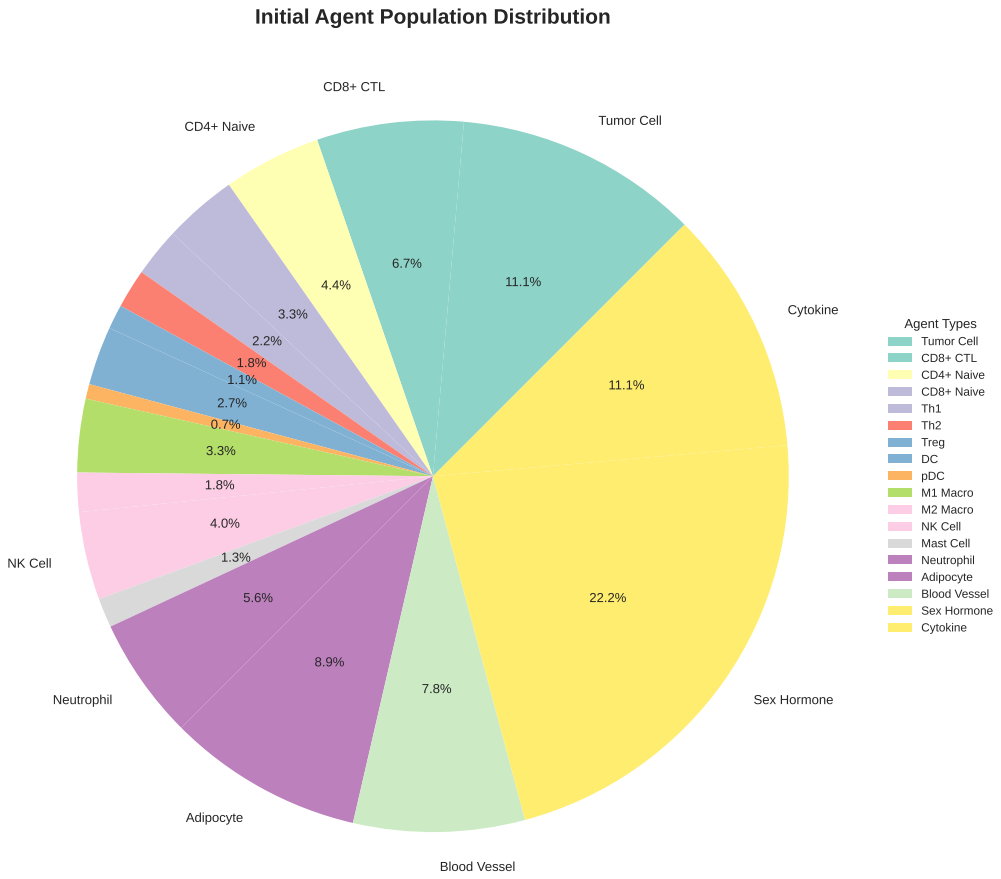


Figure 3.1: Initial agent type distribution in the RCC model. The pie chart shows the relative proportions of the 18 agent types at simulation initialization, with tumor cells forming a small but critical fraction of the total population. Immune cells dominate the initial composition, reflecting the host's surveillance capacity before tumor progression.

Table 3.1: Agent types in the RCC model.

ID	Agent Type	Class Name	Category
0	Tumor Cell	`TumorCell`	Neoplastic
1	Cytotoxic T Cell	`CytotoxicTCell`	Adaptive Immune
2	$CD8^+$ Naive T Cell	`CD8NaiveTCell`	Adaptive Immune
3	$CD4^+$ Naive T Cell	`CD4NaiveTCell`	Adaptive Immune
4	$CD4^+$ Th1 Cell	`CD4Helper1TCell`	Adaptive Immune
5	$CD4^+$ Th2 Cell	`CD4Helper2TCell`	Adaptive Immune
6	Regulatory T Cell	`TregCell`	Adaptive Immune
7	Dendritic Cell	`DendriticCell`	Innate Immune (APC)
8	Plasmacytoid DC	`PlasmacitoidDendriticCell`	Innate Immune (APC)
9	M1 Macrophage	`MacrophageM1`	Innate Immune
10	M2 Macrophage	`MacrophageM2`	Innate Immune
11	Natural Killer	`NaturalKiller`	Innate Immune
12	Mast Cell	`MastCell`	Innate Immune
13	Neutrophil	`Neutrophil`	Innate Immune
14	Adipocyte	`Adipocyte`	Stromal
15	Blood Vessel	`Blood`	Vasculature

3.1.3 Time Step Structure

Each simulation step follows a fixed sequence of operations, executed by the `RCCModel.step()` method. The ordering is designed to ensure that effects are collected before agents act, that treatment is applied at the correct time, and that spatial updates are deferred to avoid order-dependent artifacts.

1. **Collect data.** Record current population counts, glucose statistics, and kill tallies for all agent types.
2. **Check termination.** Evaluate terminal conditions (Section 3.6). If any condition is met, write the data log and halt.
3. **Apply effects.** For each agent that `needs_effect()`, collect effect objects from neighboring agents within the effect radius and sum them into the agent's `experienced_effe`
4. **Treatment.** If the current step $t \geq t_{\text{start}}$, apply all active drug effects (ICI, TKI, or both).
5. **Sex drift.** Apply the sex-dependent drift to the $CD8^+$ naive T cell population (duplication for males, removal for females), decrementing the remaining drift steps.
6. **Glucose field step.** Blood vessels inject glucose, diffusion spreads it through the 3D field, and natural decay reduces concentrations.
7. **Manage blood.** Evaluate angiogenesis conditions and create new tumor-generated blood vessels if warranted. Refresh blood access for all tumor cells via BFS from blood-connected sources.
8. **Manage search dimension.** Update the global cell search dimension based on the current tumor cell count, ensuring that immune cells can detect tumors at appropriate ranges.
9. **Shuffle and step all agents.** Take a snapshot of the agent list, shuffle it randomly to eliminate activation order bias, and invoke each agent's `step()` method. Agents may set their `_desired_pos` during this phase but are not yet moved.
10. **Deferred move.** Batch-move all agents whose `_desired_pos` differs from their current position, updating the spatial index.
11. **Spawn hormones.** Generate sex hormone agents at blood vessel locations according to sex-specific hormone chances.

3.2 Agent Hierarchy

The agent class hierarchy reflects the biological taxonomy of the cell types and the shared functionality among related agent types.

3.2.1 Base Cell Class

The Cell class provides common functionality for all cellular agents, including position tracking, deferred movement, effect collection, duplication, and neighborhood search methods.

3.2.2 TCell Base Class

The TCell class extends Cell with T lymphocyte-specific functionality: hormone perception and decay, TCR receptor matching using Hamming distance, and sex hormone modulation affecting 17 cellular behaviors.

3.2.3 Concrete Agent Classes

The 18 agent types inherit from TCell (5 T-cell types), Cell (11 cellular types), or Agent directly (2 signaling molecules).

Figure 3.2 illustrates the complete agent class hierarchy, showing the inheritance relationships and categorization of the 18 agent types.

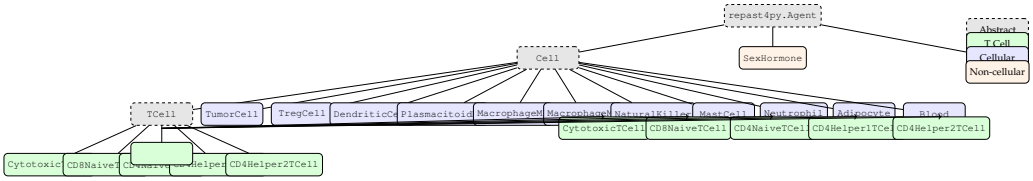


Figure 3.2: Agent class hierarchy in the RCC model. Abstract base classes are shown with dashed borders, while concrete implementations are categorized by functional type: T cells (green), other cellular agents (blue), and non-cellular signaling agents (orange).

3.3 Effect System

3.3.1 Effect Fields

The effect system is the primary mechanism for paracrine signaling between agents. Each `Cell` agent maintains an `experienced_effects` object of type `Effect` (defined in `src/systems/effect.py`), which contains 14 floating-point fields representing the accumulated influence of neighboring cells:

1. `t_kill_rate_effect` — modulates cytotoxic T cell and NK cell kill probability.
2. `t_proliferation_effect` — modulates T cell proliferation rate.
3. `t_apoptosis_effect` — modulates T cell apoptosis probability.
4. `t_activation_effect` — modulates T cell activation level (affects kill rate and proliferation multiplicatively).
5. `th1_proliferation_effect` — specifically modulates Th1 cell proliferation.
6. `th2_proliferation_effect` — specifically modulates Th2 cell proliferation.

7. `treg_differentiation_effect` — governs $CD4^+$ naive → Treg differentiation probability.
8. `macrophage_m1_mutation_effect` — governs M1 → M2 macrophage polarization probability.
9. `macrophage_m2_mutation_effect` — governs M2 → M1 macrophage repolarization probability.
10. `angiogenesis_effect` — governs the probability of new blood vessel formation.
11. `tumour_growth_effect` — modulates tumor cell duplication probability.
12. `tumour_apoptosis_effect` — modulates tumor cell apoptosis probability.
13. `dc_phagocitosis_effect` — modulates dendritic cell and pDC phagocytosis probability.
14. `nkl_kill_rate_effect` — modulates NK cell kill probability.

3.3.2 BMI Normalization

Three of the 14 effect fields are initialized with BMI-dependent baseline values rather than zero:

- `treg_differentiation_effect`: initialized to $-\hat{B} \times w_{\text{BMI_on_treg_diff}}$
- `macrophage_m1_mutation_effect`: initialized to $-\hat{B} \times w_{\text{BMI_on_m1_mutation}}$
- `macrophage_m2_mutation_effect`: initialized to $+\hat{B} \times w_{\text{BMI_on_m2_mutation}}$
- `nkl_kill_rate_effect`: initialized to $-\hat{B} \times w_{\text{BMI_nkl_kill_rate}}$

where the normalized BMI is:

$$\hat{B} = \frac{\text{BMI} - \mu_{\text{BMI}}}{\sigma_{\text{BMI}}} \tag{3.1}$$

with sex-specific parameters $\mu_{\text{BMI}} = 26.5$, $\sigma_{\text{BMI}} = 5.5$ for females and $\mu_{\text{BMI}} = 27.5$, $\sigma_{\text{BMI}} = 4.5$ for males. This normalization ensures that a BMI above the population mean increases pro-tumorigenic effects (M2 polarization, Treg differentiation) and decreases anti-tumorigenic effects (NK killing, M1 stability), consistent with clinical observations linking obesity to poorer RCC outcomes.

3.3.3 Effect Collection and Aggregation

At each simulation step, before agents execute their individual behaviors, the model's `apply_effects()` method iterates over all agents. For each agent that returns `True` from `needs_effect()`, the `collect_and_apply_effects()` method:

1. Queries the Moore neighborhood (radius 1) via the spatial index.
2. For each neighbor that is a `Cell` instance and returns `True` from `has_effect()`, retrieves its `Effect` object via `get_effect()`.
3. Aggregates all collected effects with the agent's current `experienced_effects` via the `sum_with()` method, which performs element-wise addition across all 14 fields.

The `Effect.EFFECT_RADIUS` class variable is set to 3, although the actual collection radius in the current implementation is 1 (Moore neighborhood).

3.4 Agent Interactions

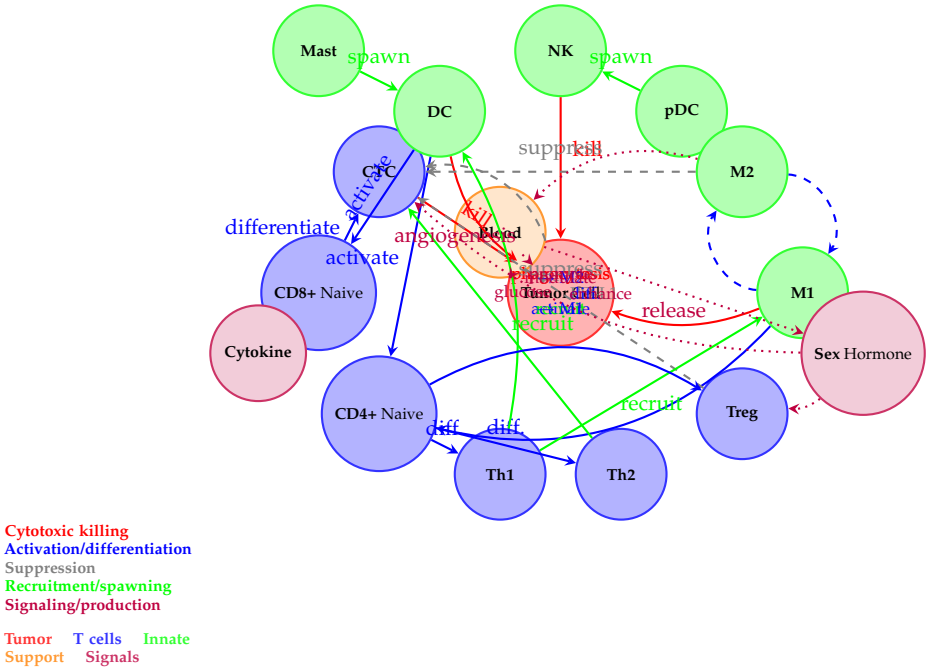


Figure 3.3: Network of agent-agent interactions in the tumor microenvironment

Table 3.2 summarizes the key interactions between agent types. Each row describes an agent type's effects on other agents, organized by the mechanism of interaction.

Table 3.2: Summary of agent interactions in the RCC model.

Agent Type	Mechanism	Effects on Other Agents
TumorCell	Growth & evasion	Proliferate, evade via PD-L1, consume glucose
CytotoxicTCell	Killing	Kill tumors, exhaustion, hormone-modulated
MacrophageM1	Anti-tumor	Phagocytosis, antigen presentation, M1→M2
MacrophageM2	Pro-tumor	Growth promotion, angiogenesis, M2→M1
TregCell	Suppression	Inhibit T cells, promote tumor
DendriticCell	Antigen presentation	Phagocytosis, T cell activation
PlasmacitoidDC	Specialized APC	Multiple effects, spawn NK cells
NaturalKiller	Innate killing	Kill tumors, glucose-dependent
MastCell	Context-dependent	Stochastic pro/anti-tumor effects
CD4NaiveTCell	Precursor	Differentiate to Th1/Th2/Treg

Continued on next page

Agent Type	Mechanism	Effects on Other Agents
CD8NaiveTCell	Precursor	Differentiate to CTL
CD4Helper1TCell	Pro-inflammatory	Recruit M1/DC, proliferate
CD4Helper2TCell	Mixed effects	Recruit immune cells, mild pro-tumor
Blood	Nutrient supply	Glucose injection, angiogenesis
SexHormone	Hormone delivery	Modulate immune responses
Neutrophil	Innate immunity	Population marker
Adipocyte	Stromal support	Structural component
Fibroblast	Stromal support	Structural component

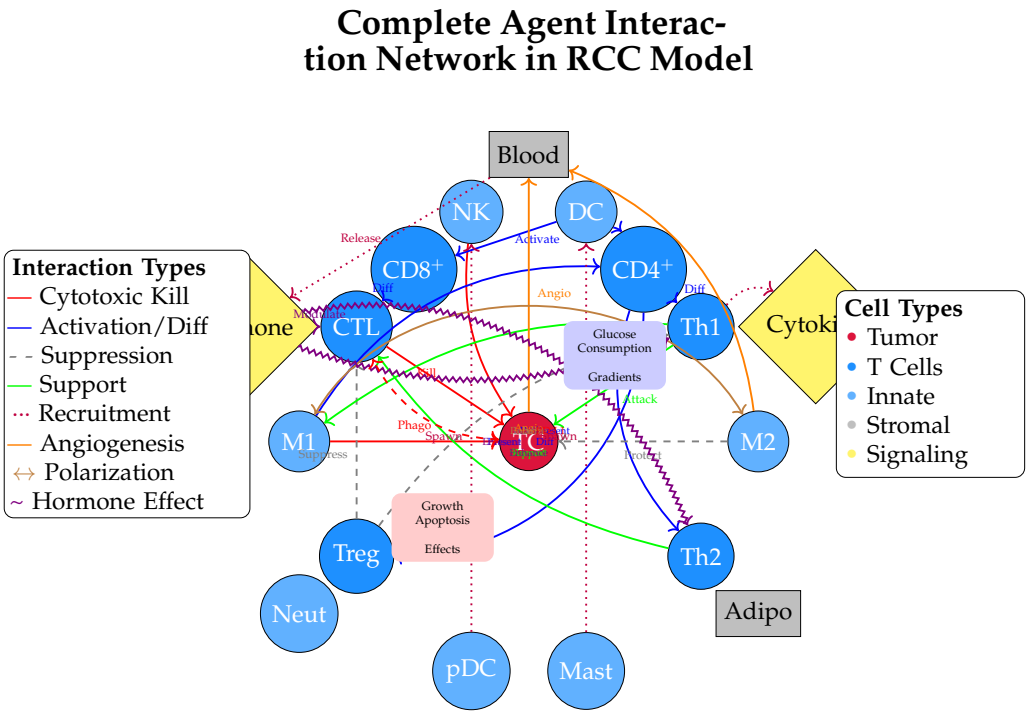


Figure 3.4: Detailed interaction network between all 18 agent types

Figure 3.5 provides a visual overview of the key interactions between agent types, showing the complex network of cytotoxic killing, activation pathways, suppression mechanisms, and recruitment cascades that govern tumor microenvironment dynamics.

[Agent interaction diagram showing tumor-immune interactions, effect propagation, and spatial relationships]

Figure 3.5: Agent interaction network in the RCC model. Arrows indicate the direction and type of interaction: red for cytotoxic killing, blue for activation/differentiation, gray dashed for suppression, green for recruitment/spawning, and purple dotted for signaling/production. Agent types are color-coded by functional category.

3.5 Movement

3.5.1 Deferred Movement Pattern

To prevent order-dependent artifacts—where an agent that moves early in the step occupies a position that a later agent expects to be empty—the model uses a deferred movement pattern:

1. During their `step()` execution, agents call `self.move(new_pos)`, which sets `self._desired_pos = new_pos` without modifying the spatial index.
2. After all agents have stepped, the model's `move_all_agents()` method iterates over all living `Cell` agents and, for each agent whose `_desired_pos` differs from `pos`, updates the spatial index and grid position.

This two-phase approach ensures that all agents make movement decisions based on the same spatial snapshot, and it eliminates bias arising from the random activation order.

3.5.2 Directed Movement: `move_towards()`

The primary movement mechanism for most agents is `move_towards(target_agent_type_id, look_up_size, distance)`, which:

1. Retrieves all living agents of the target type.
2. Finds the closest target by Manhattan distance:

$$d_M(\mathbf{a}, \mathbf{b}) = |a_x - b_x| + |a_y - b_y| + |a_z - b_z| \tag{3.2}$$

3. If the closest target is within `look_up_size`, computes the Euclidean direction vector $\mathbf{d} = \mathbf{p}_{\text{target}} - \mathbf{p}_{\text{self}}$, normalizes it, and scales by `distance`.
4. Rounds each component and clamps to grid boundaries (sticky borders).
5. Sets the resulting position as `_desired_pos`.

The `look_up_size` parameter is typically derived from the model's `cell_search_dimensio` which grows with the tumor cell count to allow immune cells to detect tumors at greater distances as the tumor mass expands.

3.5.3 Glucose Chemotaxis

When an immune cell (CTL, M1 macrophage, DC, or NK cell) fails to find a direct target via `move_towards()`, it may fall back to glucose chemotaxis. With probability `w_glucose_chemotaxis_strength`, the agent queries the glucose field for the discretized gradient at its position:

$$(\Delta x, \Delta y, \Delta z) = \text{sign}\left(\nabla C_{\text{glucose}}(\mathbf{x})\right) \tag{3.3}$$

where the gradient is computed via central finite differences with Neumann (zero-flux) boundary conditions. The agent then sets its desired position to $\mathbf{x} + (\Delta x, \Delta y, \Delta z)$, clamped to grid bounds. This mechanism causes immune cells to migrate toward glucose-rich regions—typically areas near blood vessels and actively growing tumor masses—providing a biologically plausible navigation strategy in the absence of direct antigen detection.

3.5.4 Sex Hormone Diffusion

Sex hormone agents follow a distinct movement model. At each step, a hormone agent randomly selects one of two diffusion strategies (each with 50% probability):

- **Biased diffusion.** The hormone evaluates all 26 Moore neighbors, computes the Manhattan distance from each to the nearest blood vessel, and selects a target position via weighted random choice. The weighting biases movement away from blood vessels (where hormones are spawned), promoting spatial distribution through the TME. The bias strength is controlled by a `drift_strength` parameter set to 0.25.
- **Simple diffusion.** The hormone selects a random move from a predefined set of directions biased toward positive y and z coordinates, or from the two axis-aligned directions $(0, 0, 1)$ and $(0, 1, 0)$. If the resulting position falls outside the grid, the hormone is removed from the simulation.

3.6 Terminal Conditions

The simulation terminates when any of the following conditions is met. The `terminal_condition` method returns a tuple `(terminated, survival)` indicating whether the simulation should stop and whether the outcome represents patient survival (tumor elimination).

1. **Maximum steps reached.** If $t \geq t_{\text{max}}$ (`max_steps`), the simulation terminates with `survival = False`. This represents the end of the observation period without definitive tumor clearance.
2. **Tumor elimination.** If the tumor cell count drops to zero before reaching `max_steps`, the simulation terminates with `survival = True`. This outcome indicates that the immune system (with or without treatment) successfully eliminated all tumor cells.
3. **Tumor cell count exceeds threshold.** If the number of living tumor cells reaches or exceeds `max_tumor_cells` (default: 2000), the simulation terminates with `survival = False`. This condition represents uncontrolled tumor growth that has exceeded the model's capacity and, clinically, would correspond to a large, aggressive tumor mass.

4. **Mean tumor growth rate exceeds threshold.** The mean tumor growth rate across all living tumor cells is computed as:

$$\bar{G} = \frac{1}{N_T} \sum_{i=1}^{N_T} \sigma\big(g_i + a_i,\ k = 10\big) \tag{3.4}$$

where N_T is the tumor cell count, g_i is the growth potential (combining `tumour_growth_effe` DNA-derived proliferation chance, blood access, and angiogenesis), a_i is the apoptosis-related component, and $\sigma(\cdot)$ is the sigmoid function with steepness $k = 10$. If $\bar{G} \geq w_{\text{tumour_growth_threshold}}$, the simulation terminates with `survival = False`. This condition detects cases where individual tumor growth rates are high even if the absolute cell count has not yet reached the threshold, capturing scenarios of rapidly accelerating tumor progression.

These terminal conditions collectively ensure that simulations reach a definitive outcome—either tumor elimination or tumor dominance—within a bounded number of steps, while capturing both the absolute size and the growth dynamics of the tumor population.

3.7 Immune Response Cascade Overview

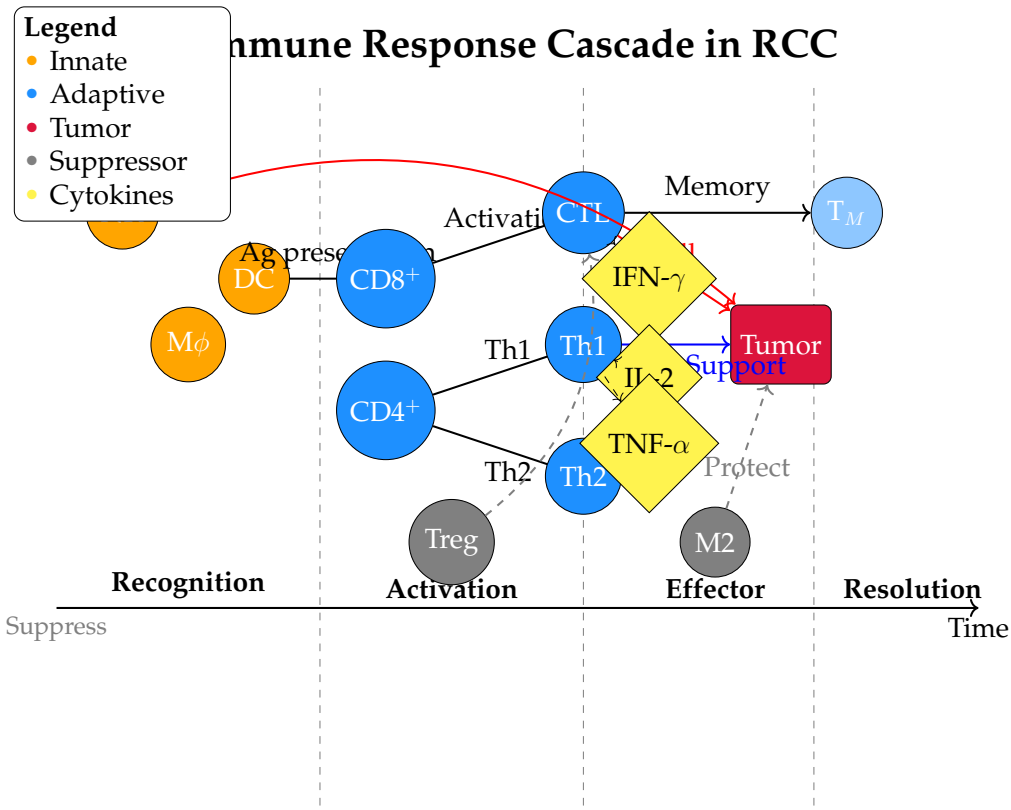


Figure 3.6: Immune response cascade from tumor detection to elimination

Figure 3.7 provides a conceptual overview of the immune response cascade as modeled in the RCC simulation. The sequence begins with tumor cell recognition and

culminates in either successful tumor elimination or immune evasion, illustrating the complex interplay between innate immunity, adaptive immunity, and tumor-mediated immunosuppression.

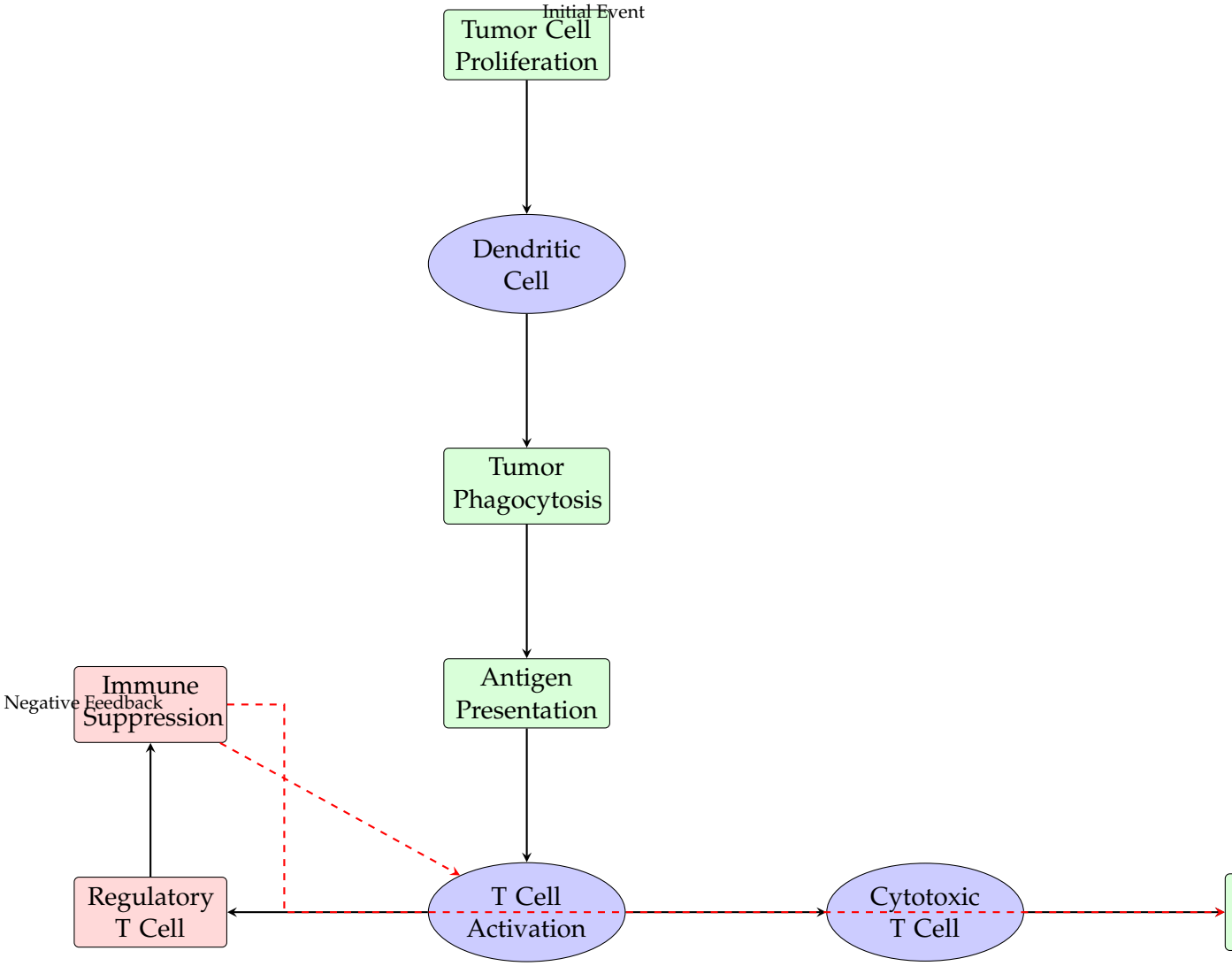


Figure 3.7: BPMN-style Immune Response Cascade in the RCC model. The process flow illustrates the sequential steps from tumor cell detection through antigen presentation to effector T cell activation and tumor elimination. Regulatory T cells provide negative feedback that can suppress both T cell activation and effector functions, creating the potential for immune evasion. Red dashed arrows indicate immunosuppressive pathways that counteract the primary immune response.

With the cellular cast assembled, we now turn to the stage itself — the metabolic landscape that determines which cells thrive and which starve. The next chapter reveals how a simple glucose field becomes the hidden battlefield where cancer and immunity wage their deadliest war.

Chapter 4

Glucose Microenvironment System

Understanding how tumors starve the immune system is central to designing effective therapies. This chapter introduces the glucose field that models this metabolic warfare — and reveals a critical tipping point that determines whether the immune system wins or loses.

This chapter describes the glucose concentration field that overlays the discrete agent grid, modeling metabolic dynamics within the tumor microenvironment. The system captures diffusion, source injection from blood vessels, consumption by tumor and immune cells, and gradient-based chemotaxis for immune cell navigation.

4.1 Biological Motivation

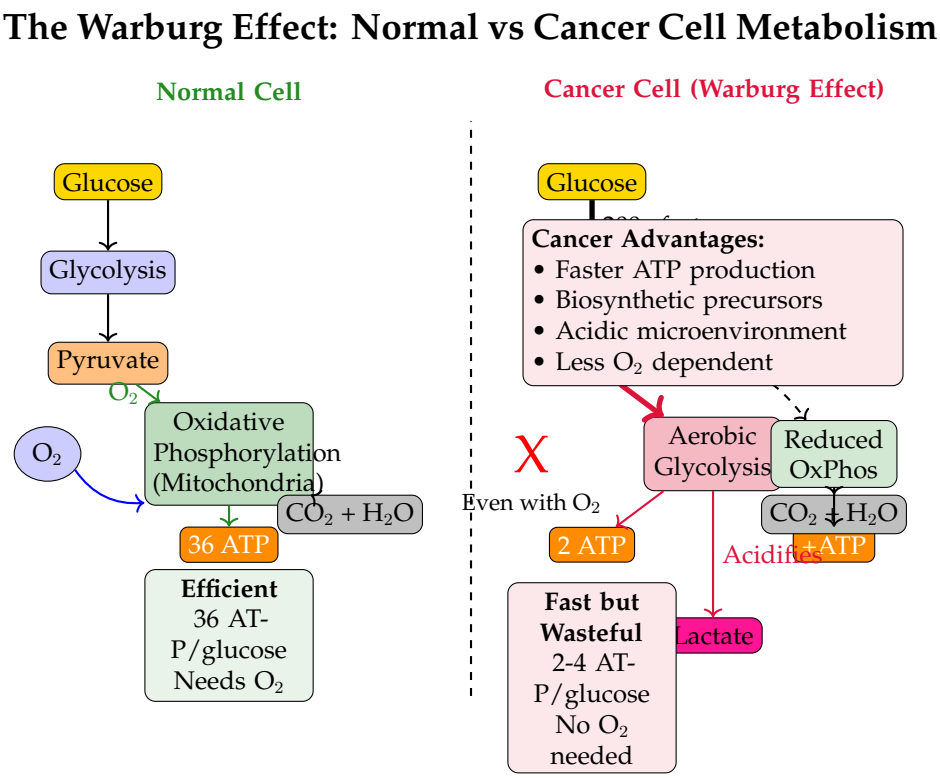


Figure 4.1: The Warburg effect: metabolic competition in the tumor microenvironment

The metabolic landscape of the tumor microenvironment plays a decisive role in determining immune-tumor interactions. Otto Warburg observed that tumor cells preferentially metabolize glucose through aerobic glycolysis, resulting in dramatically elevated glucose uptake (up to 200-fold compared to normal tissue). This metabolic competition between tumor cells and infiltrating immune cells creates nutrient gradients

that shape spatial organization, as regions surrounding tumor masses become nutrient-deprived. T cells and NK cells critically depend on glucose availability for cytotoxic activity — when glucose is scarce, T cell proliferation is impaired, IFN-γ production decreases, and cytotoxic granule release is diminished. In the absence of direct cell-cell contact, immune cells can follow glucose concentration gradients as a proxy for proximity to functional vasculature and viable tissue, providing a chemotactic fallback behavior.

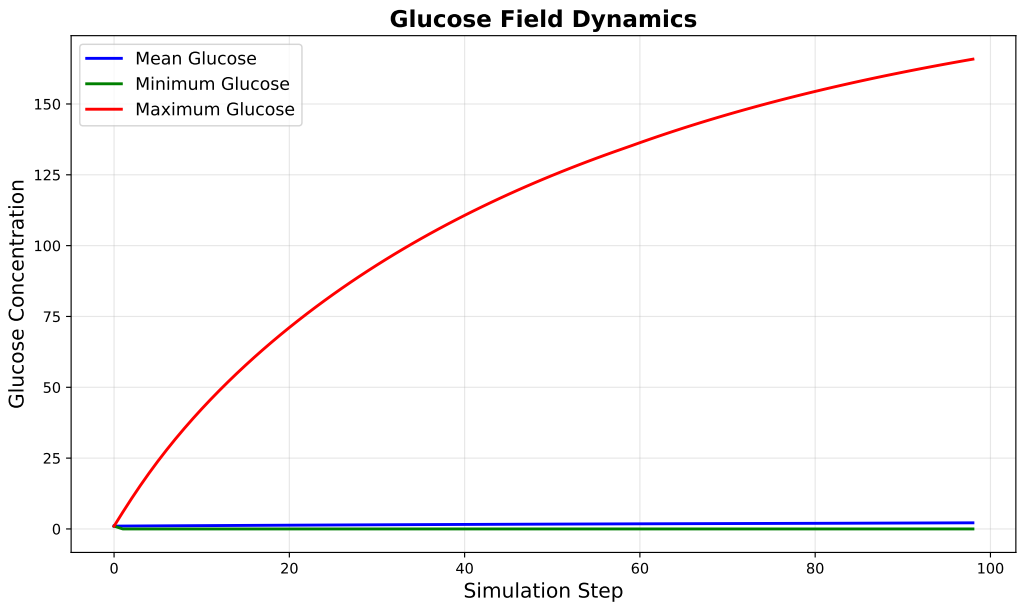


Figure 4.2: Glucose field dynamics over time showing the evolution of glucose concentration patterns in response to tumor growth and treatment. The spatial distribution reveals zones of depletion around tumor masses and gradients extending from blood vessel sources.

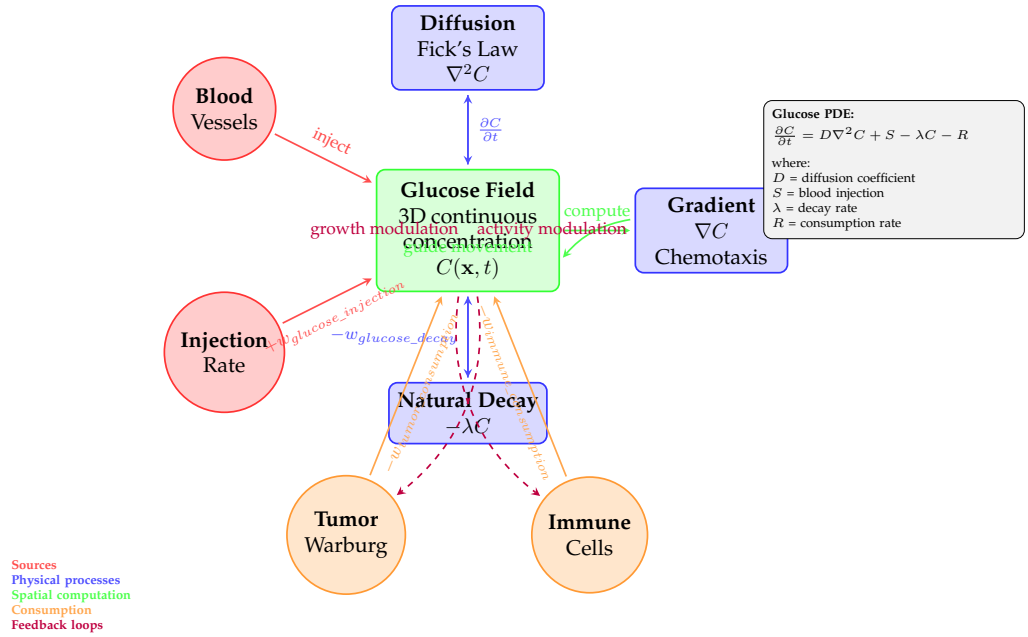


Figure 4.3: Glucose system architecture showing the diffusion field, sources from blood vessels, and consumption by cellular agents.

4.2 Mathematical Formulation

The glucose concentration field $C(\mathbf{x}, t)$ evolves according to a reaction–diffusion equation derived from Fick's second law:

$$\frac{\partial C}{\partial t} = D\,\nabla^2 C \;-\; \lambda\, C \;+\; S(\mathbf{x}) \;-\; Q(\mathbf{x}), \tag{4.1}$$

where $C(\mathbf{x}, t)$ is the glucose concentration, D is the diffusion coefficient (`w_glucose_diffusion`, default 0.1), λ is the natural decay rate (`w_glucose_decay`, default 0.01), $S(\mathbf{x})$ is the source term (glucose injection from blood vessels), and $Q(\mathbf{x})$ is the consumption term (uptake by tumor and immune cells).

The continuous Laplacian $\nabla^2 C$ is discretized on a regular 3D Cartesian grid using the standard 6-connected stencil, implemented as a 3D convolution using `scipy.ndimage.convol` The diffusion equation is advanced using explicit Euler integration with stability condition $D < 1/6$. Neumann (zero-flux) boundary conditions are enforced, corresponding to no glucose flux through simulation domain walls.

4.2.1 Source and Consumption Models

Blood vessel agents inject glucose at their grid positions at rate r_{source} (`w_glucose_source_rate`, default 5.0) per step. Tumor cells consume glucose reflecting the Warburg effect at rate r_{tumor} (`w_glucose_tumor_consumption`, default 2.0), while immune cells consume at lower rate r_{immune} (`w_glucose_immune_consumption`, default 0.5).

Tumor cell proliferation probability is modulated by glucose availability: $f_{\text{glucose}} = \min(1.0, C(\mathbf{x}_t) \cdot w_{\text{growth_sensitivity}})$, and immune cell kill rates are similarly scaled: $f_{\text{immune}} = \min(1.0, C(\mathbf{x}_i) \cdot w_{\text{immune_sensitivity}})$. When glucose is abundant ($C \geq 1.0$), activities proceed at full rate; as glucose drops below 1.0, functions are proportionally suppressed.

4.3 Gradient and Chemotaxis

The glucose gradient is computed using central finite differences, and for agent movement on the integer grid, the continuous gradient is converted to discrete step directions using the sign function. Immune cells that implement glucose chemotaxis (`CytotoxicTCell`, `MacrophageM1`, `DendriticCell`, `NaturalKiller`) employ a fallback strategy where they first attempt to move towards nearby tumor cells, and if no target is found, they move in the direction of increasing glucose concentration with probability p_{chemo} (`w_glucose_chemotaxis_strength`, default 0.3).

4.4 Implementation

The glucose system is encapsulated in the `GlucoseField` class. The glucose field is stored as a 3D NumPy array with `float64` precision, initialized uniformly to 1.0. Each simulation step executes three phases:

1. **Injection**: Blood vessels inject glucose at their positions
2. **Diffusion**: Apply discrete Laplacian convolution and Euler integration
3. **Decay**: Multiply field by $(1 - \lambda)$

Agent-level consumption occurs within each agent's `step()` method. The implementation uses vectorized NumPy operations for field updates while maintaining individual agent-level consumption behavior. The class exposes aggregate statistics (mean, total, min, max concentrations) for tracking metabolic dynamics over time.

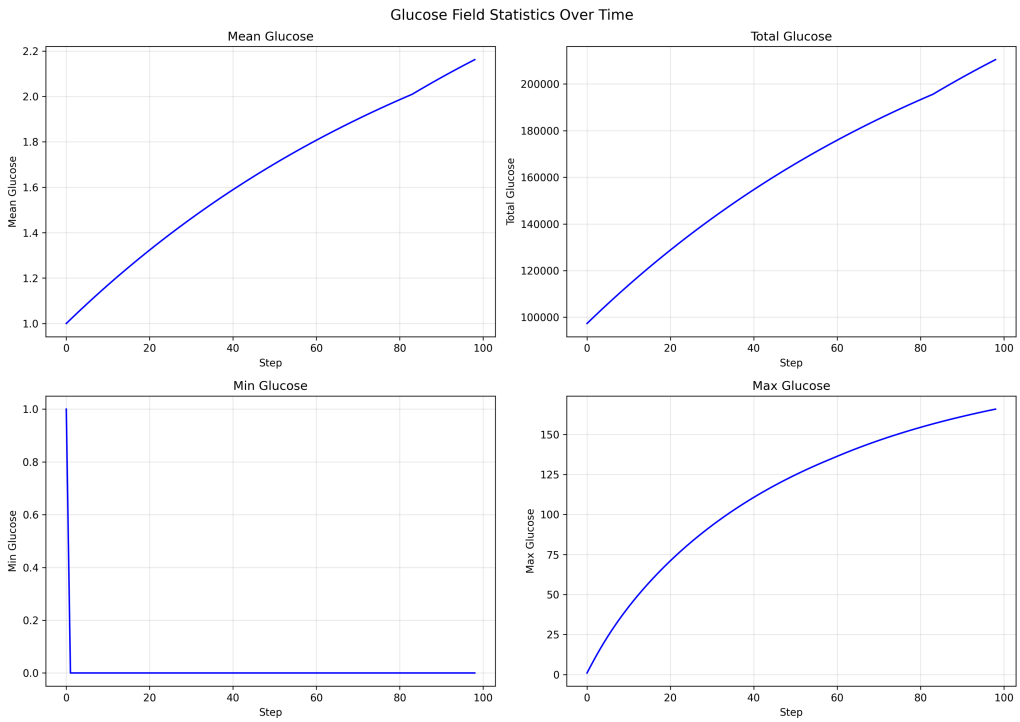


Figure 4.4: Glucose concentration time series showing the temporal evolution of mean, minimum, and maximum glucose levels throughout the simulation. The dynamics reflect the balance between blood vessel injection, cellular consumption, and spatial diffusion.

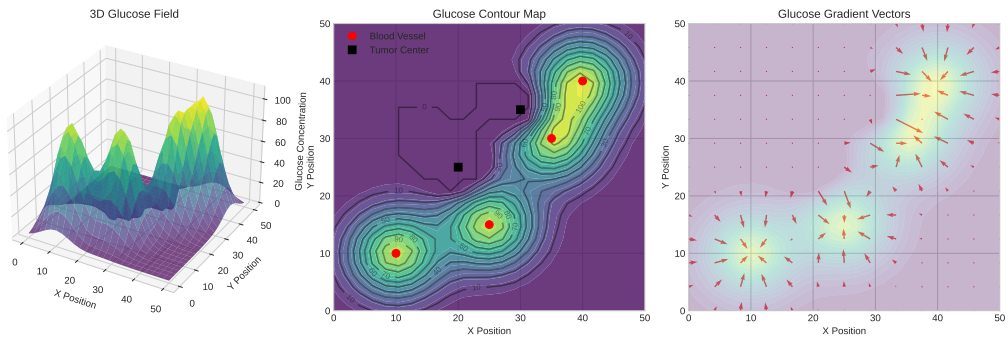


Figure 4.5: Three-dimensional visualization of the glucose concentration field showing spatial heterogeneity within the tumor microenvironment. High-concentration regions (yellow) correspond to blood vessel locations, while low-concentration zones (blue) indicate areas of intensive cellular consumption.

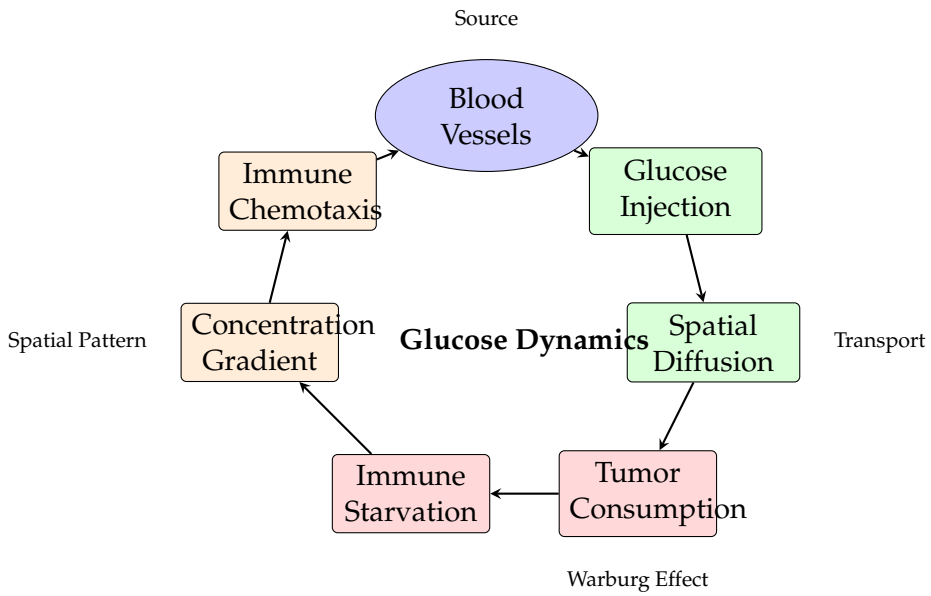


Figure 4.6: Glucose System Cycle illustrating the circular flow of glucose dynamics in the tumor microenvironment.

Chapter 5

DNA and Mutation System

Every cancer cell is genetically unique, and this diversity determines which treatments succeed or fail. This chapter reveals how 16 key genes create tumor fingerprints — and shows why males start with three times more cancer-driving mutations than females.

This chapter describes the gene-level representation used by tumor cells in the model. The DNA system encodes 16 cancer-relevant genes, supports codon-level translation, introduces mutations during cell division with genomic-instability-dependent rates, and derives phenotypic parameters that govern tumor behavior and immune evasion.

5.1 Gene Model

Each tumor cell carries a DNA object encoding 16 genes chosen for their relevance to renal cell carcinoma immunology, spanning antigen presentation, immune checkpoint signaling, DNA repair, tumor suppression, and oncogenic proliferation.

Table 5.1: The 16 genes modeled in the DNA system and their functional roles.

Gene	Category	Functional Role
MHC	Antigen Presentation	Major histocompatibility complex; presents peptide antigens on the cell surface for T cell recognition.
B2M	Antigen Presentation	Beta-2-microglobulin; essential component of MHC class I molecules.
TAP1	Antigen Presentation	Transporter associated with antigen processing 1; loads peptides onto MHC I.
TAP2	Antigen Presentation	Transporter associated with antigen processing 2; works with TAP1 for peptide loading.
JAK1	Antigen Presentation	Janus kinase 1; IFN-γ signaling for MHC upregulation.
JAK2	Antigen Presentation	Janus kinase 2; cooperates with JAK1 in interferon signaling.
CD274	Immune Checkpoint	PD-L1; binds PD-1 on T cells to induce immune checkpoint inhibition.

Continued on next page

Gene	Category	Functional Role
BRCA1	DNA Repair	Breast cancer type 1 susceptibility protein; key DNA double-strand break repair gene controlling genomic stability.
TP53	Tumor Suppression	Tumor protein p53; "guardian of the genome," triggers apoptosis in damaged cells.
RB1	Tumor Suppression	Retinoblastoma protein; cell cycle checkpoint regulator.
CDKN2A	Tumor Suppression	Cyclin-dependent kinase inhibitor 2A (p16); inhibits CDK4/6 to arrest cell cycle.
MYC	Oncogene	Transcription factor promoting cell growth and proliferation.
RAS	Oncogene	GTPase signal transducer in the MAPK proliferation pathway.
EGFR	Oncogene	Epidermal growth factor receptor; receptor tyrosine kinase driving proliferation.
HER2	Oncogene	Human epidermal growth factor receptor 2; amplified in aggressive tumors.
APC	Tumor Suppression	Adenomatous polyposis coli; Wnt pathway regulator, loss promotes proliferation.

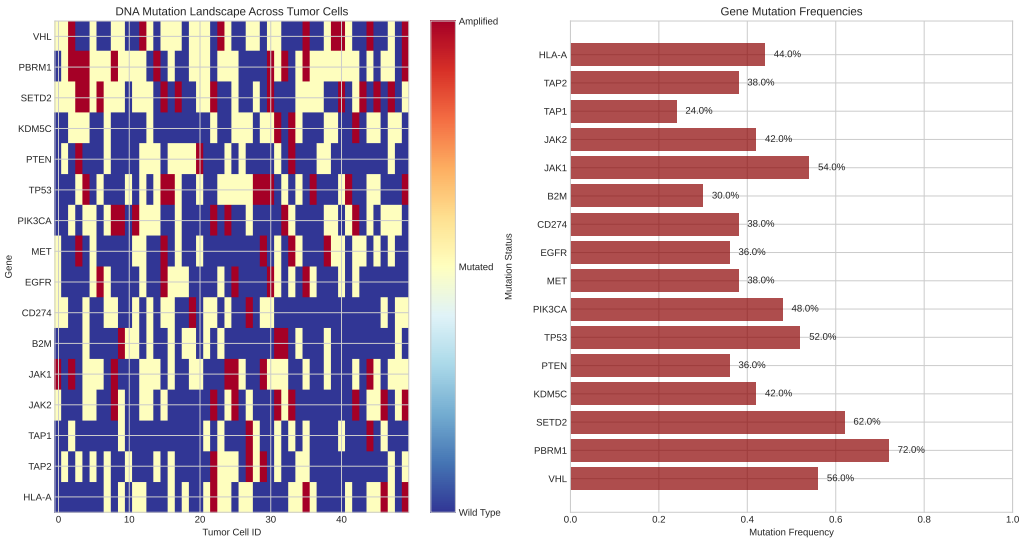


Figure 5.1: DNA mutation heatmap across the 16 modeled genes showing the distribution of mutations in the tumor cell population. Each row represents a gene and each column represents a tumor cell lineage. Color intensity indicates the severity of mutation (darker colors represent higher mutation scores). The heterogeneous mutation patterns drive clonal diversity and immune evasion.

Each gene occupies a defined region within a synthetic DNA sequence assembled by concatenating random spacer, promoter region (including TATA box), coding region with codon-encoded wildtype protein, and stop codon. The model implements full 64-codon table mapping for nucleotide triplet to amino acid translation.

5.2 Mutation Mechanics

Mutations accumulate through injected mutations at initialization and stochastic point mutations during DNA duplication. Genomic instability is controlled by BRCA1 mutation status, with mutation rate scaling as $\mu = (s_{BRCA1})^{\max(1,e_{BRCA1})} \times 0.05$. Initial mutations are sex-dependent: males start with 3 BRCA1 mutations, females with 1, establishing different baseline genomic instability patterns. Point mutations occur during DNA duplication at rate μ per nucleotide, introducing base substitutions that may alter codon identity and protein sequence.

5.3 Gene Expression and Phenotypic Effects

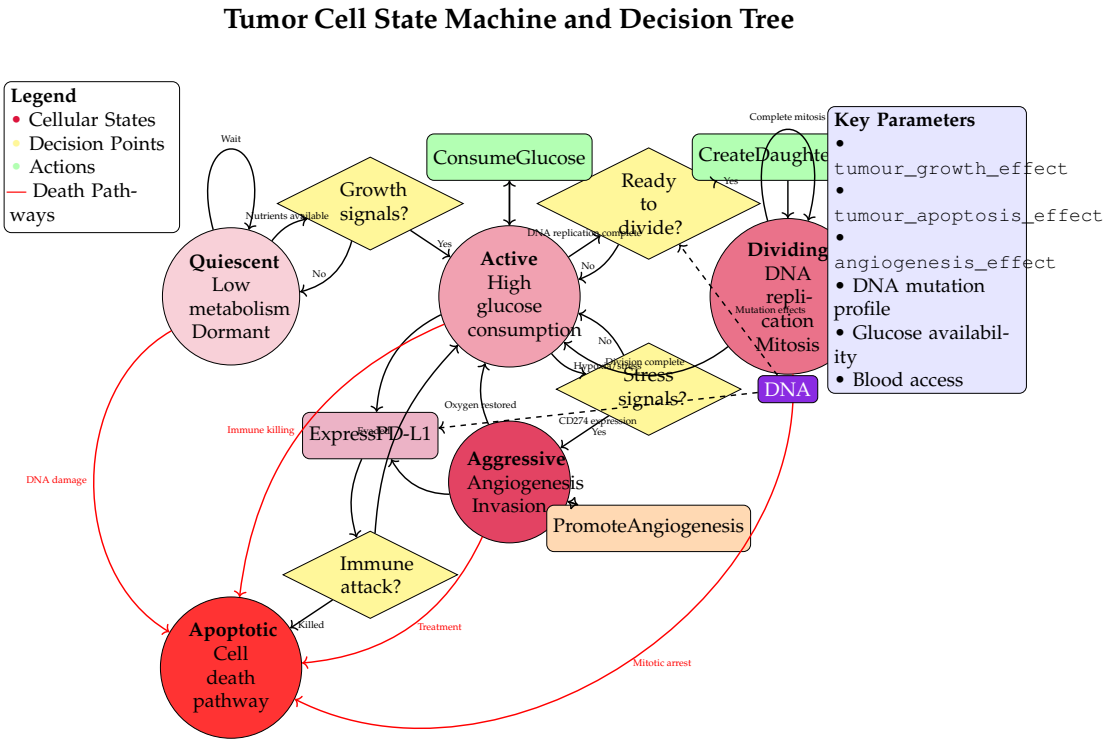


Figure 5.2: State machine diagram of tumor cell lifecycle

Each gene's phenotypic contribution is determined by promoter expression (TATA motif count) and mutation score (Levenshtein distance from wildtype protein). The DNA system computes five aggregate phenotypic parameters:

Antigen Presentation Chance: $p_{AP} = \max(0, 6 - \frac{\sum_{g \in \{MHC,JAK1,JAK2,B2M,TAP1,TAP2\}} e_g \cdot s_g}{6})$ represents the tumor cell's ability to present antigens, with mutations reducing this

probability for immune evasion.

Checkpoint Inhibition Chance: $p_{CPI} = e_{CD274} \cdot s_{CD274}$ determined by PD-L1 gene, with higher mutated expression increasing T cell checkpoint engagement probability.

Genomic Instability: Controls mutation rate during DNA duplication based on BRCA1 dysfunction level.

Tumor Suppression Chance: $p_{TS} = \frac{1}{3} \sum_{g \in \{TP53, RB1, CDKN2A\}} e_g \cdot (1 - s_g)$ with functional tumor suppressors increasing apoptosis probability.

Extra Proliferation Chance: $p_{EP} = \frac{1}{5}(\sum_{g \in \{MYC, RAS, EGFR, HER2\}} e_g \cdot (1 - s_g) + e_{APC} \cdot s_{APC})$ capturing oncogene-driven growth.

5.4 Neoantigen Generation

Neoantigens are 10-mer subsequences extracted from each translated mutated protein, representing molecular targets for T cell recognition. Self-antigens are similarly computed from wildtype sequences to establish immune tolerance. T cell receptors recognize neoantigens through Hamming distance-based matching between antigen hashes and receptor values, with successful recognition requiring non-self status and distance within threshold.

These genetic mutations create vulnerabilities that treatments can exploit. The next chapter examines exactly how immune checkpoint inhibitors and targeted therapies turn cancer's own molecular weaknesses against itself.

Chapter 6

Treatment System

These genetic mutations create vulnerabilities that treatments can exploit. This chapter examines exactly how immune checkpoint inhibitors and targeted therapies work — and reveals why the ICI+TKI combination is more than the sum of its parts.

This chapter describes the pharmacological treatment layer of the model, which simulates Immune Checkpoint Inhibitors (ICI), Tyrosine Kinase Inhibitors (TKI), and their combination. The treatment system is activated after a configurable number of simulation steps and modulates tumor and immune cell behaviors through drug-specific mechanisms.

6.1 Drug Architecture

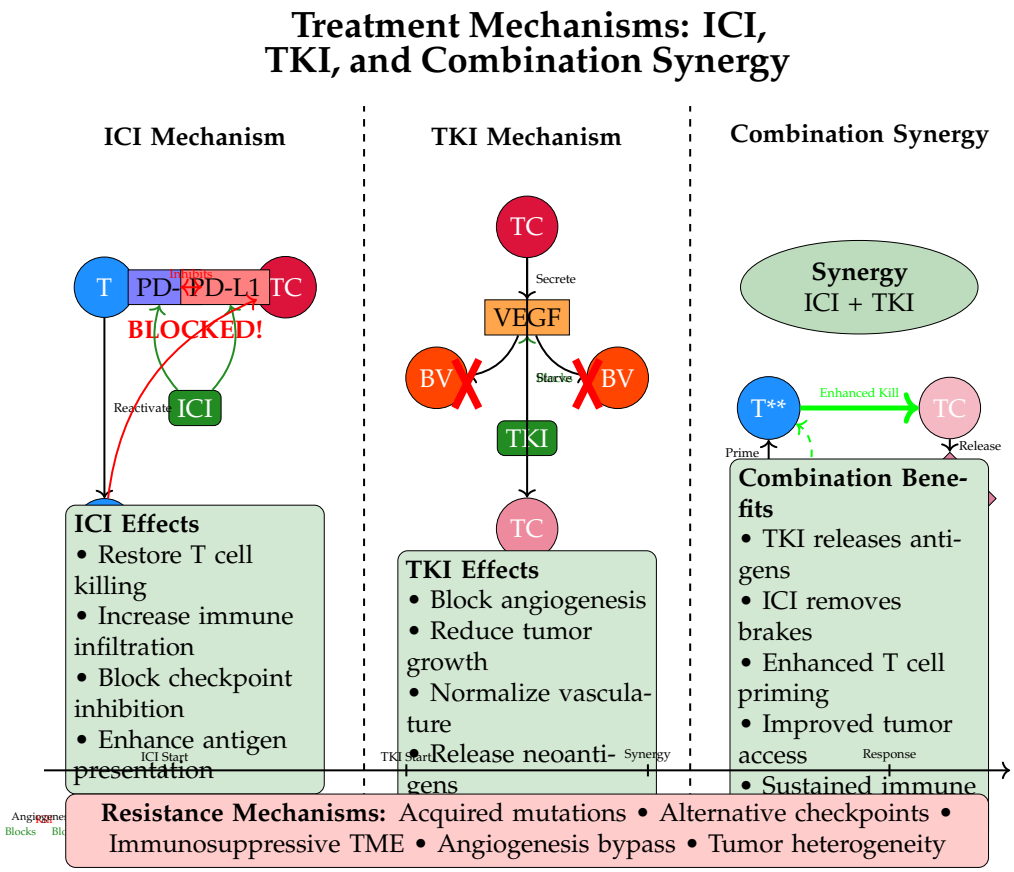


Figure 6.1: ICI and TKI treatment mechanisms and their targets

The treatment system follows a composition-based design where individual drugs inherit from an abstract Drug base class with step() method and proportion parame-

ter controlling treatment allocation fraction. The Treatment class distributes capacity equally among drugs: single drug therapy receives proportion = 1.0, combination therapy gives each drug 0.5, modeling clinical dose adjustments. Treatment activates after treatment$_{s}tartsteps(default:100), establishingbaselinediseasestatebeforeintervention.$

Available treatment configurations via treatment patient parameter:

- "None" – no drugs (control)
- "ICI" – immune checkpoint inhibitor monotherapy
- "TKI" – tyrosine kinase inhibitor monotherapy
- "ICI+TKI" – combination therapy

6.2 Immune Checkpoint Inhibitors (ICI)

PD-1/PD-L1 checkpoint interaction suppresses T cell cytotoxicity, and ICI antibodies block this pathway to restore T cell function. ICIDrug implements three mechanisms with probability proportional to drug effectiveness: (1) PD-L1 blockade on tumor cells, disabling checkpoint inhibition by setting ICI$_{e}ffectflagtoTrue; (2)Tcellactivationrestorationacrossa$
$L1+tumorencounters; and(3)immuneinfiltrationenhancementbyincrementingimmune_{i}nfiltration_{f}$

6.3 >

Tyrosine Kinase Inhibitors (TKI)

VEGF drives angiogenesis in RCC due to frequent VHL gene inactivation, making anti-angiogenic TKI therapy (axitinib, sunitinib) a cornerstone treatment by blocking VEGF receptor signaling. TKIDrug implements four effects with probability proportional to effectiveness: (1) tumor growth suppression by reducing tumour$_{g}rowth_{e}ffect, (2)angiogene$
$naiveTcells, shiftingimmunebalancetowardanti-tumoractivity.$

6.4 Combination Therapy

ICI+TKI combination therapy provides each drug proportion = 0.5, reflecting clinical dose adjustments. The combination exploits synergistic mechanisms: TKI normalizes aberrant tumor vasculature improving immune cell infiltration, ICI restores cytotoxic capacity of infiltrating T cells, TKI-induced immunogenic cell death releases neoantigens while ICI ensures activated T cells remain functional, and TKI suppresses immunosuppressive Tregs while ICI restores effector T cell activity. Clinical trials (KEYNOTE-426, JAVELIN Renal 101) demonstrated combination superiority over monotherapy, with ARON-1 examining sex-specific response patterns.

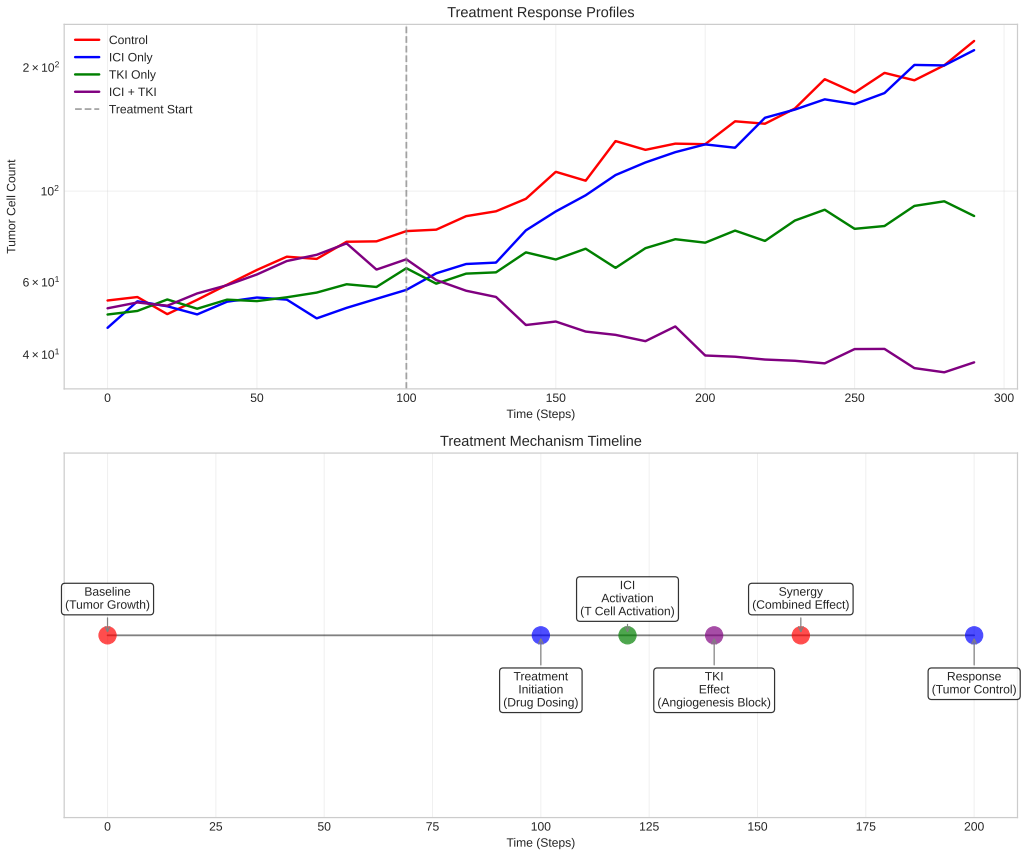


Figure 6.2: Treatment timeline showing the sequential application of ICI and TKI therapies across simulation steps. The figure illustrates drug administration schedules, onset of action, and temporal coordination in combination therapy protocols.

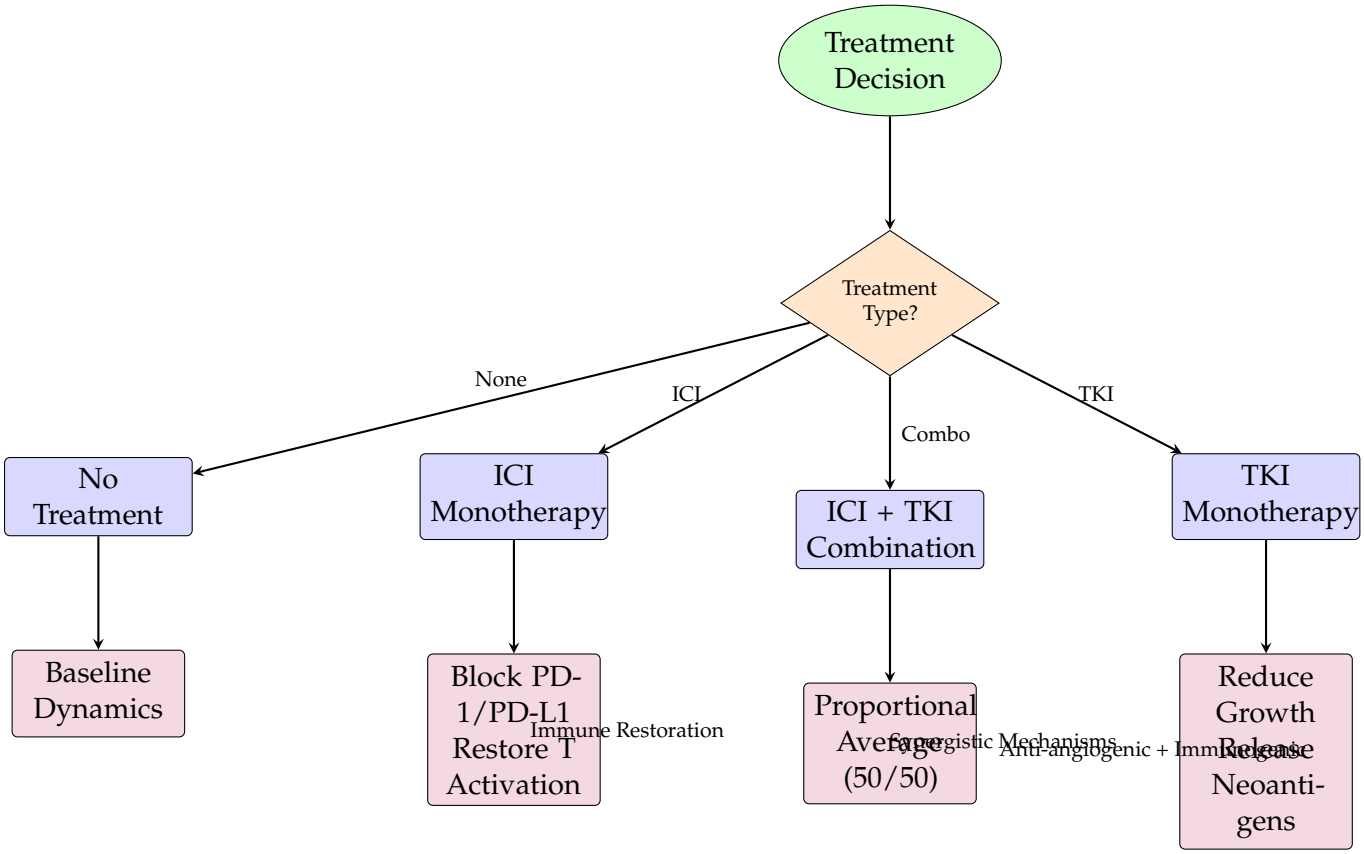


Figure 6.3: Treatment Decision Flow showing the four treatment modalities available in the RCC model.

Chapter 7

Sex Hormone System

Why do females respond better to some cancer treatments while males benefit more from others? This chapter reveals how sex hormones orchestrate immune responses — and shows how estrogen at moderate levels actually boosts the immune system, explaining the female advantage in cancer survival.

This chapter describes the sex hormone subsystem, which models the immunomodulatory effects of estrogen, testosterone, and progesterone on the tumor microenvironment. Sex-specific differences in hormone profiles produce distinct immune dynamics, contributing to the well-documented sex disparity in RCC incidence and treatment response.

7.1 Biological Basis

Renal cell carcinoma exhibits a pronounced sex disparity with males affected approximately twice as often as females, yet treatment efficacy varies significantly between sexes. Estrogen exerts complex, concentration-dependent effects: at low to moderate levels, it promotes Th1-type immune responses, enhances IFN-γ production, and stimulates T cell activation; at high concentrations, it shifts towards Th2-type responses and increases inhibitory checkpoint engagement. Testosterone is generally immunosuppressive at elevated levels, reducing Th1 proliferation, decreasing IFN-γ production, and promoting immune checkpoint engagement. Progesterone has minimal effects at low levels, mildly enhances checkpoint inhibition at moderate levels, and strongly suppresses Th1 proliferation while promoting Th2 differentiation at high concentrations. These clinical observations suggest that sex hormones modulate the immune response in ways that affect both tumor susceptibility and treatment efficacy.

7.2 Hormone Agent Model

Sex hormones are modeled as discrete agents (estrogen, testosterone, progesterone) that diffuse through the simulation grid, interact with nearby immune cells, and decay over time. The `SexHormone` class inherits from `repast4py.core.Agent` and is registered as agent type ID 16. Hormones are spawned from blood vessel positions with sex-specific probabilities (male: estrogen 0.2, progesterone 0.1, testosterone 1.0; female: estrogen 0.9, progesterone 0.8, testosterone 0.2). Once spawned, hormones perform biased random walk through the grid with drift away from blood vessels, and those exiting grid boundaries are removed, providing natural hormone clearance with effective lifespan of approximately 50 steps.

Table 7.1: Hormone spawn probabilities by patient sex.

Sex	Estrogen	Progesterone	Testosterone
Male	0.2	0.1	1.0
Female	0.9	0.8	0.2

7.3 Hormone Effects on T Cells

T cells detect sex hormones via `perceive_sex_hormone()` method, maintaining dictionaries of perceived hormone levels that decay by 10% per step. The `CellHormonalInfluenceE` enumeration defines 17 effect channels through which hormones modulate T cell behavior:

Table 7.2: Hormonal influence effect channels.

Effect	Description
KILL_RATE	Cytotoxic kill rate modulation.
PROLIFERATION	General proliferation rate.
APOPTOSIS	Programmed cell death susceptibility.
ACTIVATION	T cell activation state.
DIFFERENTIATION	Differentiation probability (e.g., naive to effector).
TH1_PROLIFERATION	Th1-specific proliferation.
TH2_PROLIFERATION	Th2-specific proliferation.
IL_4	Interleukin-4 signaling effect.
IL_6	Interleukin-6 signaling effect.
IL_13	Interleukin-13 signaling effect.
IFN_GAMMA	Interferon-gamma signaling effect.
TNF_ALPHA	Tumor necrosis factor alpha effect.
MHC_I_UPREGULATION	MHC class I expression modulation.
MHC_II_UPREGULATION	MHC class II expression modulation.
COSTIMULATION_CD28	CD28 costimulatory signal strength.
CTLA4_SUPPRESSION	CTLA-4 inhibitory checkpoint engagement.
PD1_INHIBITION	PD-1 inhibitory checkpoint engagement.

Perceived hormone levels are normalized to [0,1] scale and mapped to concentration regimes (low < 0.33, medium 0.33-0.66, high ≥ 0.66). Effects are weighted by hormone-specific importance: $w_{\text{estrogen}} = 0.45$, $w_{\text{testosterone}} = 0.30$, $w_{\text{progesterone}} = 0.25$.

7.4 Sex Drift

The model implements population-level sex drift that adjusts CD8+ naive T cell populations based on patient sex, reflecting testosterone's stimulation of CD8+ T cell production in the thymus. The drift operates as gradual population adjustment over 15 steps with maximum drift $d_{\max} = 0.25$: $\delta = d_{\max}/n$ per step, where direction $\sigma = +1$ for males (population increase) and $\sigma = -1$ for females (population decrease). Step factor $f_{\text{step}} = 1 + \sigma \cdot \delta$ determines duplication probability (males) or removal probability (females), producing approximately ±25

$$\delta = \frac{d_{\max}}{n}, \qquad f_{\text{step}} = 1 + \sigma \cdot \delta \tag{7.1}$$

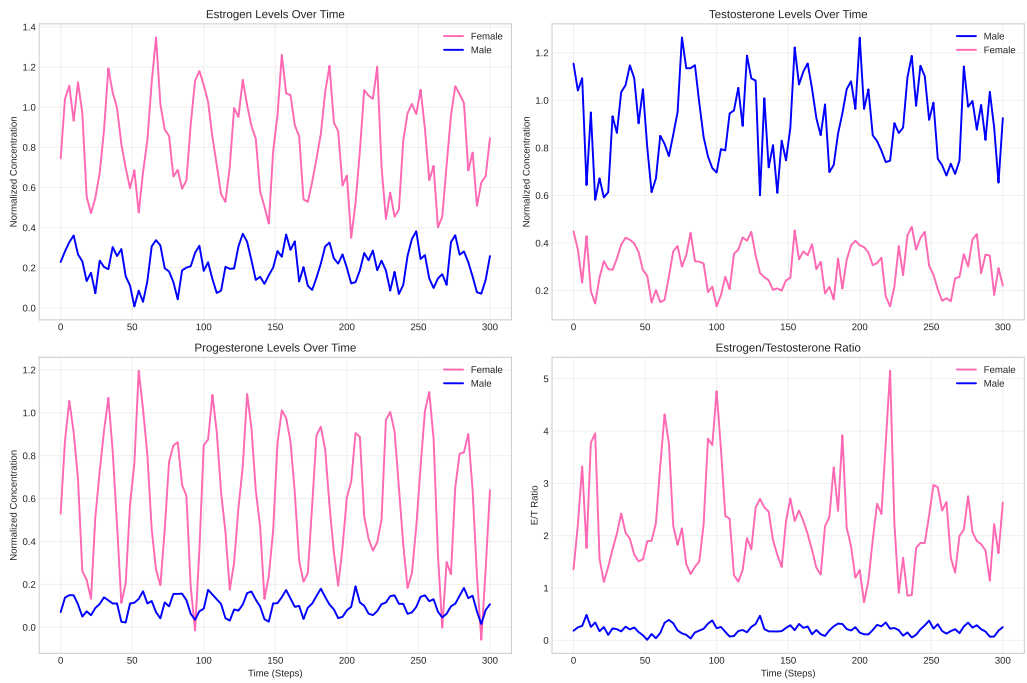


Figure 7.1: Sex hormone dynamics over time showing the evolution of estrogen, testosterone, and progesterone levels in the tumor microenvironment. The hormone concentrations fluctuate based on sex-specific spawning rates, cellular consumption, and spatial diffusion patterns.

Chapter 8

Implementation

Converting biological complexity into working code requires more than good programming — it demands architectural vision. This chapter reveals how moving from Mesa to Repast4Py enabled true 3D simulation and MPI parallelism, transforming a proof-of-concept into a production-scale virtual tumor.

This chapter examines the architectural decisions, design patterns, and implementation strategies that transform biological theory into executable software. Rather than cataloging code structure, we focus on the engineering problems encountered and the design patterns employed to solve them, revealing how computational constraints shape biological model design and vice versa.

8.1 Framework Migration: Mesa to Repast4Py

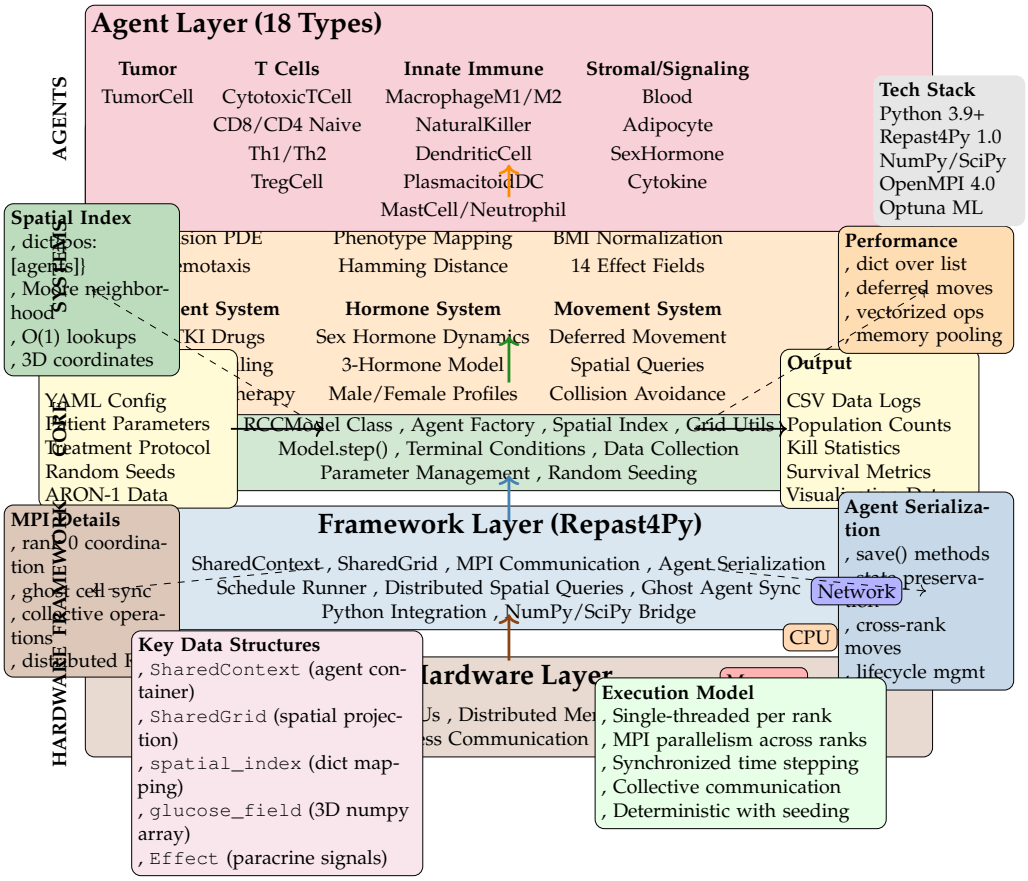


Figure 8.1: Software architecture overview of the RCC simulation system

Table **??** presents a comprehensive comparison between Mesa and Repast4Py implementations, highlighting the key architectural differences and enhancements made during the porting process.

The implementation uses Command pattern for deferred movement, Observer pattern for data collection, and Strategy pattern for treatment modalities.

8.2 Simulation Orchestration: The Main Event Loop

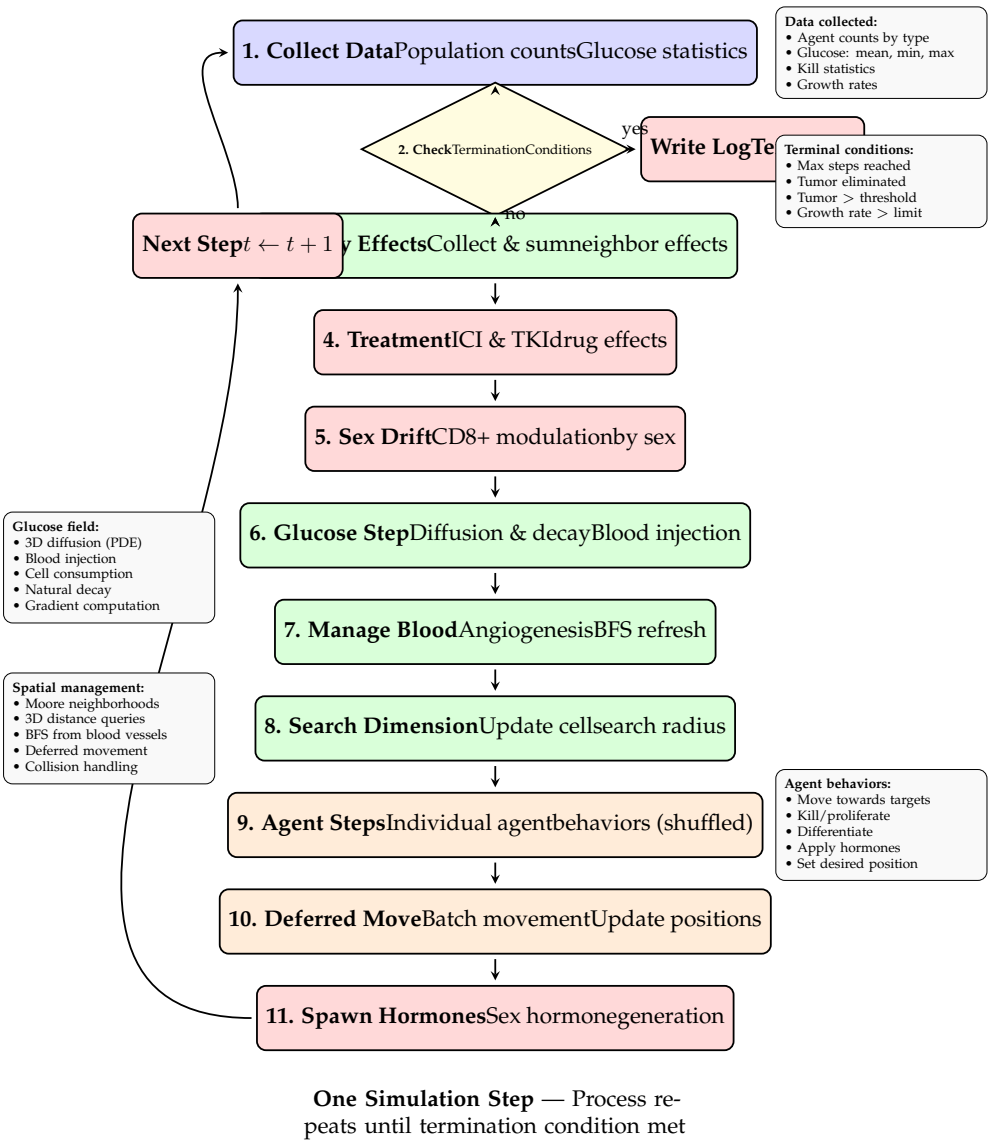
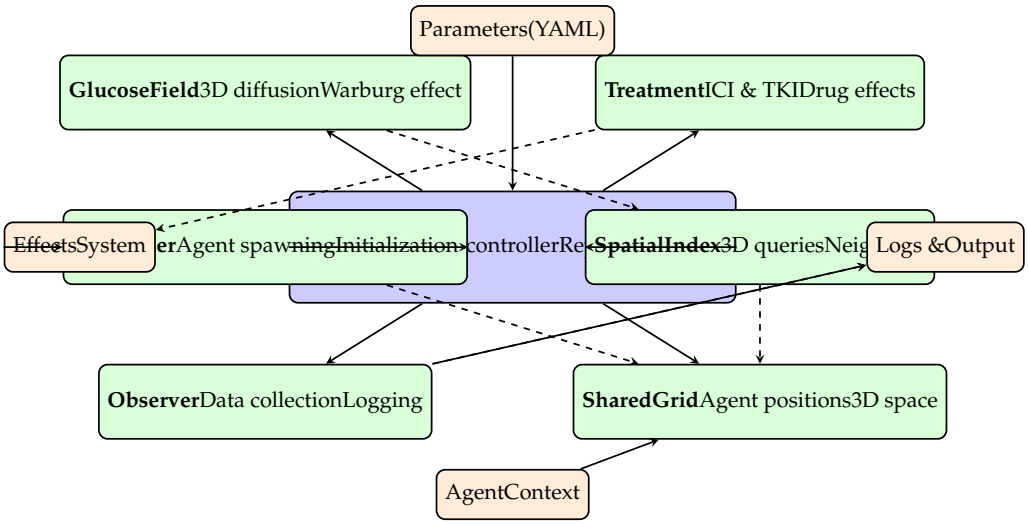


Figure 8.2: Simulation execution pipeline

The main simulation loop executes a fixed sequence: data collection, termination check, effect application, treatment, glucose field updates, agent stepping, and deferred movement execution. This ordering respects biological causality while preventing execution order bias.

The codebase is organized into agent classes (17 cell types), system modules (glucose field, DNA mutations, effect accumulation), treatment implementations, and pa-



Main Step Sequence: 1. collect_data → 2. check_termination → 3. apply_effects → 4. treatment → 5. sex_drift → 6. glucose_step → 7. manage_blood → 8. agent_steps → 9. move → 10. spawn_hormones

Figure 8.3: Model architecture showing component relationships

rameter management. Repast4Py's RCCModel class directly manages SharedContext, SharedGrid, and agent lifecycle without framework inheritance.

Figure 8.4 illustrates the overall architecture of the RCCModel, showing the main components and their data flow relationships.

[Architecture diagram showing RCCModel, GlucoseField, CellAdder, Observer, Treatment, SpatialIndex, and SharedGrid components with their interconnections]

Figure 8.4: Architecture of the RCCModel system. The main model coordinates interactions between the glucose field (3D diffusion), agent management (cell adder, spatial index, shared grid), data collection (observer), and treatment systems. The step sequence annotation shows the ordered execution of simulation phases.

8.3 Spatial Index Design

Repast4Py lacks built-in neighbor queries, so the model maintains a manual spatial index using defaultdict(list) that maps coordinates to agent lists. Grid utilities provide 3D neighborhood computation using Moore or Von Neumann neighborhoods with boundary checking.

8.4 Agent Serialization

Repast4Py agents require (id, type, rank) construction and maintain uid tuples for MPI distribution. The AgentType IntEnum maps 18 cell types to integer IDs, and each agent implements a save() method returning (uid, state$_tuple)forserialization.Typeidentificationusesintege$

8.5 Mesa to Repast4Py Porting

Table 8.1: Key API mapping from Mesa to Repast4Py.

Mesa	**Repast4Py**
`mesa.Agent(model)`	`repast4py.core.Agent(id, type, rank)`
`self.model.random`	`model.rng` (stdlib `random.Random`)
`self.model.grid.get_neighborhood()`	`grid.utils.get_neighbors_3d(spatial_index, ...)`
`isinstance(agent, TumorCell)`	`agent.uid[1] == AgentType.TUMOR_CELL`
`self.model.schedule.agents`	`model._all_agents`
`model.agents_by_type[Type]`	`model.get_agents_by_type_id(type_id)`
`TumorCell.blood_source` (class var)	`model.tumor_blood_sources`
`Cell.search_dimension` (class var)	`model.cell_search_dimension`

The porting required moving class-level state to model instances, implementing custom random number generation, and preserving deferred movement patterns to prevent execution order artifacts.

8.6 Glucose Field Implementation

The GlucoseField class uses NumPy 3D arrays and SciPy convolution for performance and control. Diffusion uses a 6-connected discrete Laplacian kernel with zero-flux boundary conditions via mode='nearest'. The implementation avoids Repast4Py's SharedValueLayer for simplicity and debugging convenience.

8.7 Entry Point and Execution

The simulation runs via 'mpirun -n 1 python run.py' using Repast4Py's schedule runner to execute the main step() loop with data collection, termination checking, effect propagation, treatment, glucose updates, agent stepping, and deferred movement.

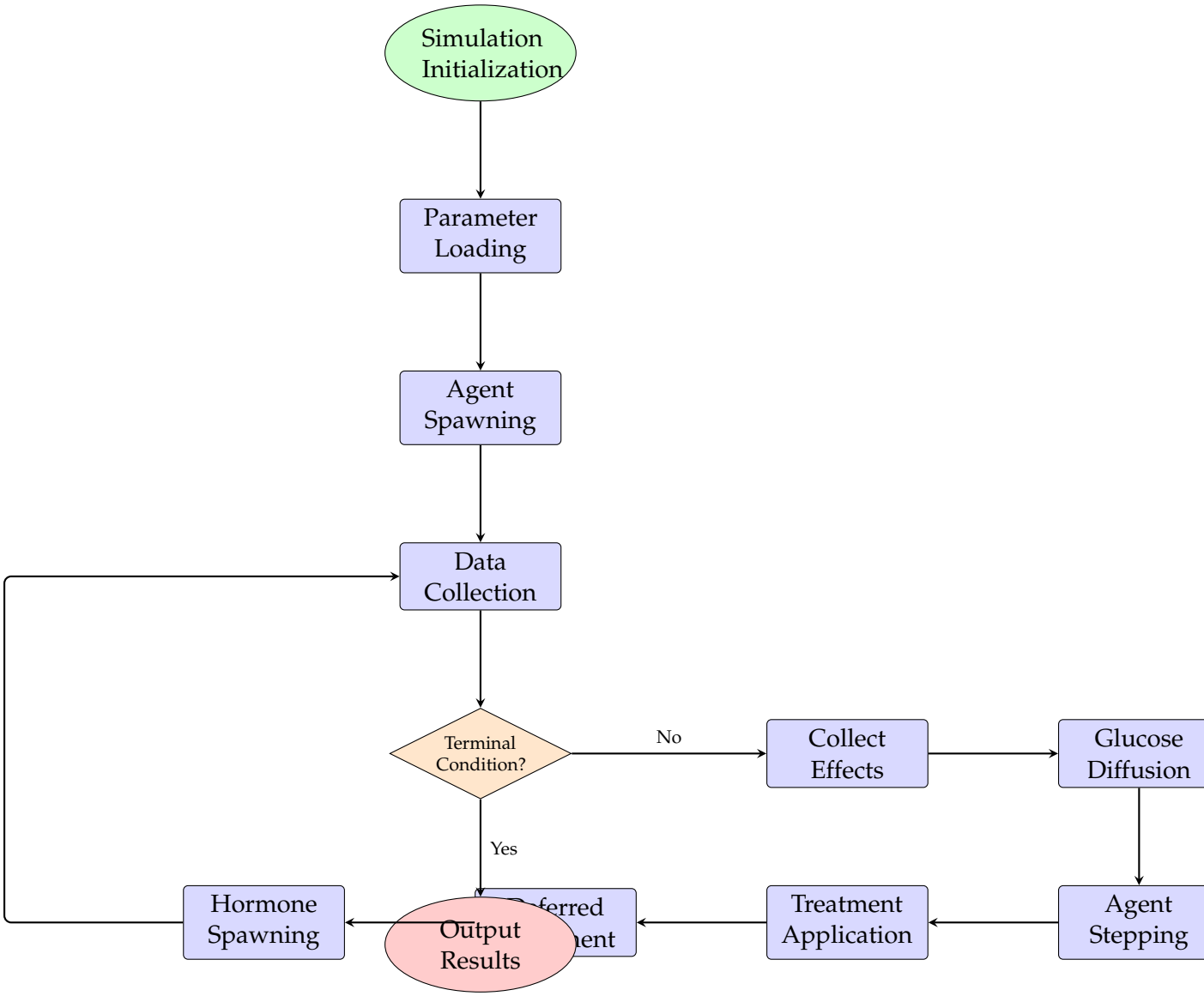


Figure 8.5: BPMN-style Simulation Lifecycle for the RCC model. The simulation follows a deterministic sequence: initialization and parameter loading, then iterative execution of the main loop. Each step collects data, checks termination conditions, applies agent effects, performs glucose diffusion, executes treatment protocols, steps agents, and resolves movement. The loop continues until a terminal condition is met, at which point results are output.

Chapter 9

Parameters and Optimization

The model employs a 114-parameter hierarchy: 4 model parameters (volume, resolution, seed), 18 patient parameters (clinical variables), and 92 weight parameters (biological process rates) optimized via Bayesian methods against clinical trial data.

9.1 Parameter Categories

Parameters are organized into three dataclass objects: ModelParameters, PatientParameters, and WeightParameters.

ModelParameters: 4 simulation configuration parameters (volume, block size, random seed, max steps). PatientParameters: 18 patient-specific parameters (sex, BMI, treatment type, start time, plus 14 immune cell concentrations). WeightParameters: 92 biological process rate parameters optimized via Bayesian methods.

9.2 Weight Parameters

The 84 weight parameters control probability thresholds and scaling factors for all agent behaviors, organized by cell type. These parameters are optimized via Bayesian methods to match clinical trial outcomes.

9.2.1 Macrophage M1 Weights (6 weights + 2 biases)

Table 9.1: Macrophage M1 weight parameters.

Parameter	Default	Description
`w_m1_mutation`	1.0	M1 macrophage mutation threshold scaling.
`b_m1_mutation`	0.0	Bias for M1 mutation threshold.
`w_m1_move`	1.0	M1 macrophage look-up movement radius scaling.
`w_m1_phagocytosis`	1.0	M1 phagocytosis chance scaling.
`w_m1_digest`	1.0	M1 digestion rate scaling.
`w_m1_t_kill_rate`	1.0	M1 effect on T cell kill rate.
`w_m1_th1_proliferation`	1.0	M1 effect on Th1 proliferation.
`w_th1_proliferation`	1.0	Th1 proliferation scaling (via M1 influence).
`b_th1_proliferation`	0.0	Bias for Th1 proliferation.

9.2.2 Macrophage M2 Weights (5 weights + 1 bias)

Table 9.2: Macrophage M2 weight parameters.

Parameter	Default	Description
`w_m2_mutation`	1.0	M2 macrophage mutation threshold scaling.
`b_m2_mutation`	0.0	Bias for M2 mutation threshold.
`w_m2_move`	1.0	M2 macrophage look-up movement radius.
`w_m2_t_kill_rate`	1.0	M2 effect on T cell kill rate.
`w_m2_tumour_growth`	1.0	M2 effect on tumour growth promotion.
`w_m2_angiogenesis`	1.0	M2 effect on angiogenesis.

9.2.3 Tumor Cell Weights (8 weights + 3 biases)

Table 9.3: Tumor cell weight parameters.

Parameter	Default	Description
`w_tumor_apoptosis_eff`	1.0	Effect-driven apoptosis probability scaling.
`w_tumor_apoptosis_dna`	1.0	DNA-driven extra suppression chance scaling.
`b_tumor_apoptosis`	0.0	Bias for apoptosis probability.
`w_tumor_growth_eff`	0.6	Effect-driven growth probability scaling.
`w_tumor_growth_dna`	1.0	DNA-driven extra proliferation chance scaling.
`b_tumor_growth`	0.0	Bias for growth probability.
`w_tumor_angiogenesis`	1.0	Angiogenesis initiation probability scaling.
`b_tumor_angiogenesis`	0.0	Bias for angiogenesis probability.
`w_antigen_presentation`	1.0	Antigen presentation chance scaling.
`b_antigen_presentation`	0.0	Bias for antigen presentation.
`w_angiogenesis_tumor_growth`	1.0	Angiogenesis contribution to growth (when blood access is present).
`w_gene_pd1_inhibition`	1.0	PD-1/PD-L1 checkpoint pathway inhibition scaling.

9.2.4 CD4+ T Cell Weights (9 weights + 3 biases)

Table 9.4: CD4+ T cell weight parameters.

Parameter	Default	Description
`w_cd4_treg_diff_effect`	1.0	Effect-driven Treg differentiation scaling.
`w_cd4_treg_diff_horm`	1.0	Hormone-driven Treg differentiation scaling.
`b_cd4_treg_diff`	0.0	Bias for Treg differentiation.
`w_cd4_th1_proliferation_effect`	1.0	Effect-driven Th1 proliferation scaling.
`w_cd4_th1_proliferation_horm`	1.0	Hormone-driven Th1 proliferation scaling.
`b_cd4_th1_proliferation`	0.0	Bias for Th1 proliferation.
`w_cd4_th1_spawn_m1`	1.0	Th1 spawning of M1 macrophages.
`w_cd4_th1_spawn_dc`	1.0	Th1 spawning of dendritic cells.
`w_cd4_th2_spawn_m1`	1.0	Th2 spawning of M1 macrophages.
`w_cd4_th2_spawn_t`	1.0	Th2 spawning of T cells.
`w_cd4_th2_proliferation_effect`	1.0	Effect-driven Th2 proliferation scaling.
`w_cd4_th2_proliferation_horm`	1.0	Hormone-driven Th2 proliferation scaling.
`b_cd4_th2_proliferation`	0.0	Bias for Th2 proliferation.

9.2.5 Cytotoxic T Cell Weights (5 weights + 2 biases)

Table 9.5: Cytotoxic T cell (CD8+) weight parameters.

Parameter	Default	Description
`w_cytotoxic_proliferation`	1.0	Proliferation probability scaling.
`w_cytotoxic_apoptosis`	1.0	Apoptosis probability scaling.
`b_cytotoxic_apoptosis`	0.0	Bias for cytotoxic T cell apoptosis.
`w_cytotoxic_kill`	1.0	Tumor cell kill probability scaling.
`b_cytotoxic_kill`	0.0	Bias for cytotoxic kill probability.
`w_cytotoxic_pd1_inhibition`	1.0	PD-1 inhibition susceptibility scaling.
`w_cytotoxic_move`	1.0	Movement radius scaling for cytotoxic T cells.

9.2.6 Mast Cell Weights (6 parameters)

Table 9.6: Mast cell weight parameters.

Parameter	Default	Description
`w_mast_cell_angiogenesis`	1.0	Mast cell contribution to angiogenesis effect.
`w_mast_cell_m1_mutation`	1.0	Mast cell effect on M1 mutation.
`w_mast_cell_t_kill_rate`	1.0	Mast cell effect on T cell kill rate.
`w_mast_cell_tumour_apoptosis`	1.0	Mast cell contribution to tumour apoptosis.
`w_mast_cell_tumour_growth`	1.0	Mast cell contribution to tumour growth.
`w_mast_cell_spawn_dc`	1.0	Mast cell probability of spawning dendritic cells.

9.2.7 NK Cell Weights (3 weights + 1 bias)

Table 9.7: Natural killer cell weight parameters.

Parameter	Default	Description
`w_natural_killer_kill_rate`	1.0	NK cell direct kill rate scaling.
`b_natural_killer_kill_rate`	0.0	Bias for NK cell kill rate.
`w_nkl_t_kill_rate`	1.0	NK cell effect on T cell kill rate amplification.

9.2.8 Plasmacytoid DC Weights (6 weights + 1 bias)

Table 9.8: Plasmacytoid dendritic cell weight parameters.

Parameter	Default	Description
`w_pdc_nkl_spawn`	1.0	pDC probability of spawning NK cells.
`b_pdc_nkl_spawn`	0.0	Bias for pDC NK cell spawning.
`w_pdc_angiogenesis`	1.0	pDC contribution to angiogenesis effect.
`w_pdc_treg_diff`	1.0	pDC effect on Treg differentiation.
`w_pdc_t_proliferation`	1.0	pDC effect on T cell proliferation.
`w_pdc_t_kill`	1.0	pDC effect on T cell kill rate.
`w_pdc_nkl_kill`	1.0	pDC effect on NK cell kill rate.

9.2.9 Treg Weights (5 parameters)

Table 9.9: Regulatory T cell weight parameters.

Parameter	Default	Description
`w_treg_t_kill_rate`	1.0	Treg suppression of T cell kill rate.
`w_treg_t_proliferation`	1.0	Treg suppression of T cell proliferation.
`w_treg_t_apoptosis`	1.0	Treg induction of T cell apoptosis.
`w_treg_activation`	1.0	Treg activation threshold scaling.
`w_treg_dc_phagocytosis`	1.0	Treg effect on DC phagocytosis.

9.2.10 Global Weights (4+ parameters)

Table 9.10: Global weight parameters.

Parameter	Default	Description
`w_search_dimension`	0.5	Controls adaptive search radius for agent interactions: $\text{dim} = \max(5, \lfloor n_{\text{tumor}}/(10 \cdot w) \rfloor)$.
`w_tumour_growth_threshold`	1.0	Terminal condition threshold for mean tumour growth rate.
`w_cell_base_death_prob`	0.009	Per-step base death probability for all cells.
`w_progressive_exhaustion`	0.05	Rate of progressive immune cell exhaustion.
`w_ici_effectiveness`	1.0	ICI drug effectiveness scaling.
`w_tki_effectiveness`	1.0	TKI drug effectiveness scaling.
`w_sex_hormone_cd8`	1.0	Sex hormone influence on CD8+ population.
`w_dc_phagocytosis_effect`	1.0	DC phagocytosis effect scaling.
`b_dc_phagocytosis`	0.0	Bias for DC phagocytosis.
`w_max_drift`	0.25	Maximum sex-specific CD8+ drift over the drift period.
`receptor_threshold_variation`	1	Integer variation in TCR matching threshold.

9.3 Glucose Parameters

The glucose microenvironment system introduces 8 new weight parameters, all prefixed with `w_glucose_`. These parameters control the physical dynamics of glucose diffusion, the metabolic consumption rates, and the coupling between glucose availability and agent behavior.

Table 9.11: Glucose weight parameters.

Parameter	Default	Description
`w_glucose_diffusion`	0.1	Diffusion coefficient D for the 3D Laplacian update. Stability requires $D < 1/6 \approx 0.167$; the default value of 0.1 provides moderate diffusion well within the stable regime.
`w_glucose_decay`	0.01	Natural glucose decay rate per step. Applied multiplicatively as $(1 - d)$ to the entire field, modeling background metabolic consumption by tissue not explicitly represented as agents.
`w_glucose_source_rate`	5.0	Amount of glucose injected per blood vessel agent per step. Blood vessels act as point sources, creating localized glucose-rich zones that subsequently diffuse outward.
`w_glucose_tumor_consumption`	2.0	Glucose consumed per tumor cell per step. Reflects the elevated glycolytic rate characteristic of the Warburg effect [10]. Tumor cells consume approximately 4× the glucose of immune cells.
`w_glucose_immune_consumption`	0.5	Glucose consumed per immune cell per step. Represents the metabolic cost of immune cell activation and effector function.
`w_glucose_growth_sensitivity`	1.0	Sensitivity of tumor growth probability to local glucose concentration. The glucose factor is computed as $\min(1, C \cdot s)$, where C is the local concentration and s is this sensitivity parameter.
`w_glucose_immune_sensitivity`	1.0	Sensitivity of immune cell effector function to local glucose concentration. Controls how strongly glucose deprivation impairs immune cell activity.

Parameter	Default	Description
`w_glucose_chemotaxis_strength`	0.3	Probability that an immune cell follows the glucose gradient (chemotaxis) rather than performing a random walk. At each step, with probability $p = 0.3$, the cell moves in the direction of increasing glucose concentration.

9.3.1 Parameter Interactions

The glucose parameters interact with the existing weight parameters in several ways:

- **Tumor growth modulation**: The tumor cell duplication probability (governed by `w_tumor_growth_eff` and `w_tumor_growth_dna`) is multiplied by a glucose factor $\min(1, C \cdot \texttt{w_glucose_growth_sensitivity})$. When glucose is abundant, this factor approaches 1 and growth proceeds at the base rate; when glucose is depleted, growth is suppressed.
- **Immune function modulation**: Immune cell kill rates and activation probabilities can be scaled by an analogous glucose factor using `w_glucose_immune_sensitivity`, coupling metabolic resource availability to immune effectiveness.
- **Spatial competition**: The balance between `w_glucose_source_rate` (supply from blood vessels) and `w_glucose_tumor_consumption` (demand from tumor cells) determines the spatial glucose profile. High tumor consumption relative to supply creates depletion zones that impair both further tumor growth and immune infiltration.

9.4 Bayesian Optimization

9.4.1 Framework

The model uses Optuna [1] for Bayesian hyperparameter optimization. Optuna's Tree-structured Parzen Estimator (TPE) sampler efficiently explores high-dimensional parameter spaces by building probabilistic models of the objective function landscape.

9.4.2 Objective Function

The optimization objective is to minimize the prediction error of the model against the ARON-1 clinical dataset. For each Optuna trial, the following process is executed:

1. A set of weight parameters is suggested by the TPE sampler within predefined bounds.

2. For each patient case in the ARON-1 dataset, a simulation is run with the patient's clinical features (sex, BMI, treatment, immune cell concentrations) and the suggested weight parameters.

3. The simulation outcome (survival/death, number of steps) is compared against the clinical outcome.

4. The per-case error is computed as:

$$\text{error}_i = \begin{cases} (\text{steps}_i - \text{OS}_i)^2 & \text{if prediction matches outcome} \\ \text{HIGH_PENALTY}^2 & \text{if prediction contradicts outcome} \end{cases} \tag{9.1}$$

   where OS_i is the observed overall survival (in steps) and HIGH_PENALTY $= 500$.

5. The trial objective value is the mean error across all cases.

9.4.3 Search Space

The search space is defined in the `suggest_parameters()` function and covers the full 92-parameter weight space:

- **Positive weights** (62 parameters): Sampled log-uniformly in $[0.25, 4.0]$. The log-uniform distribution ensures adequate exploration of both small and large scaling factors.
- **Biases** (15 parameters): Sampled uniformly in $[-1.0, 1.0]$.
- **Effectiveness weights**: `w_ici_effectiveness` and `w_tki_effectiveness` in $[0.0, 1.0]$.
- **Threshold weights**: `w_tumour_growth_threshold` in $[0.75, 0.99]$; `w_cell_base_death_p` in $[0.001, 0.1]$; `w_progressive_exhaustion` in $[0.005, 0.3]$.
- **Integer parameters**: `receptor_threshold_variation` in $\{0, 1, 2, 3\}$.
- **Glucose parameters** (8 parameters):
  - `w_glucose_diffusion` in $[0.01, 0.16]$ (respecting the $D < 1/6$ stability constraint).
  - `w_glucose_decay` in $[0.001, 0.1]$.
  - `w_glucose_source_rate` in $[1.0, 20.0]$.
  - `w_glucose_tumor_consumption` in $[0.5, 10.0]$.
  - `w_glucose_immune_consumption` in $[0.1, 2.0]$.
  - `w_glucose_growth_sensitivity` in $[0.1, 5.0]$.
  - `w_glucose_immune_sensitivity` in $[0.1, 5.0]$.
  - `w_glucose_chemotaxis_strength` in $[0.05, 0.8]$.

9.4.4 Optimization Configuration

The optimization runs for $N = 250$ trials, each evaluating every patient case in the ARON-1 dataset. Training uses `MPI.COMM_SELF` to create isolated single-rank model instances, avoiding MPI communication overhead during optimization. Trial errors and the best-so-far error are logged to `data/trial_errors.csv` for convergence analysis.

```
study = optuna.create_study(direction="minimize")
study.optimize(wrapped_objective, n_trials=250)
```

Listing 9.1: Optuna study creation and execution.

9.5 Clinical Dataset

9.5.1 ARON-1 Study

The ARON-1 study is a clinical trial of advanced renal cell carcinoma patients treated with immune checkpoint inhibitors (ICI), tyrosine kinase inhibitors (TKI), or their combination [8, 6]. The dataset provides the ground truth against which the model is calibrated.

9.5.2 Dataset Structure

The clinical dataset is stored in `src/learning/data/ARON.csv` and contains the following features for each patient case:

- **Patient features**: sex (`'F'/'M'`), BMI, treatment type (ICI, TKI, or ICI+TKI).
- **Immune cell concentrations**: 13 cell type concentrations matching the `PatientParameters` fields.
- **Outcome**: Overall survival (OS) in time units, and a binary death indicator.

The dataset is loaded with Pandas, with the treatment column parsed as a list of strings to support combination therapies:

```
def load_dataset(file_path: str) -> pd.DataFrame:
    def treatment_to_list(treatment_str):
        return [t.strip() for t in treatment_str.split('+')]
    return pd.read_csv(file_path,
                       converters={"treatment": treatment_to_list})
```

Listing 9.2: Dataset loading.

9.6 Parameter Sensitivity Analysis

Figure 9.1 demonstrates the sensitivity of simulation outcomes to key parameter variations within the defined ranges. This analysis identifies which parameters have the

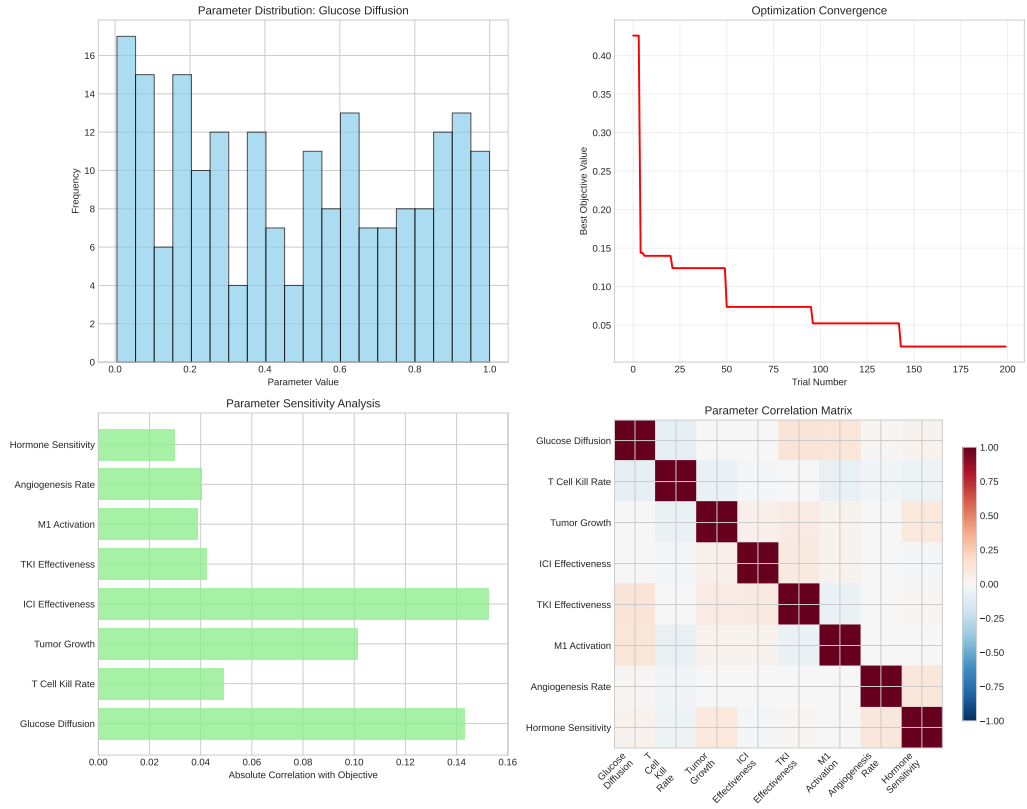


Figure 9.1: Parameter sensitivity analysis showing the impact of key model parameters on tumor cell population dynamics. The analysis varies parameters within their defined ranges while keeping others at baseline values. Most sensitive parameters include glucose consumption rates, treatment effectiveness weights, and immune cell interaction parameters. Error bars represent variance across multiple simulation runs.

strongest influence on tumor dynamics and treatment efficacy, providing guidance for parameter selection and model robustness assessment.

Each row of the dataset is used to parameterize a simulation run during optimization, with the patient's clinical features passed as keyword arguments to `RCCModel`. The simulation's survival prediction is then compared against the clinical outcome to compute the trial error.

With 92 parameters tuned against clinical reality, we can finally ask the question that motivated this entire project: does sex matter in RCC treatment? The next chapter reveals surprising answers that challenge conventional treatment wisdom.

Chapter 10

Simulation Results

With 92 parameters tuned against clinical reality, we can finally ask the question that motivated this entire project: does sex matter in kidney cancer treatment? This chapter reveals how single-seed results can be catastrophically misleading — and shows why 160 simulation runs are needed to uncover the truth.

This chapter presents the results from systematic simulation of the RCC agent-based model across six scenarios: sex-specific baseline (no treatment) conditions and three treatment modalities (ICI, TKI, and combination therapy) stratified by biological sex.

10.1 Baseline Simulation

The baseline scenarios run the simulation with default parameters and no treatment for both male and female virtual patients.

Male patients showed aggressive tumor progression, reaching 2,034 final tumor cells over 287 simulation steps, with 29 tumor cell kills and mean glucose concentration of 5.042. Growth was driven by three injected mutations, testosterone-mediated immune suppression, and extensive angiogenesis.

Female patients demonstrated significantly different tumor behavior, with only 92 final tumor cells over 299 simulation steps, 15 tumor cell kills, and mean glucose concentration of 3.167. The slower progression reflected lower initial mutational burden (1 vs. 3 mutations) and estrogen-mediated immune response enhancement.

10.2 Treatment Scenarios

Three treatment modalities were evaluated across both sexes, revealing complex interactions between therapy mechanisms and biological sex.

ICI Monotherapy: Achieved survival (0 tumor cells) in both sexes through PD-1/PD-L1 blockade. Male patients required 143 steps with 11 kills (glucose: 2.370), while females achieved faster elimination at 129 steps with 11 kills (glucose: 2.480), potentially reflecting enhanced efficacy in the estrogen-modulated microenvironment.

TKI Monotherapy: Demonstrated efficacy through anti-angiogenic effects. Males achieved survival at 222 steps with 11 kills (glucose: 2.606), while females showed faster response at 113 steps with 11 kills (glucose: 2.370), suggesting enhanced sensitivity to anti-angiogenic therapy in the estrogen-modulated environment.

ICI+TKI Combination: Produced surprising sex-specific differences. Males achieved survival at 162 steps with 11 kills (glucose: 2.562), but females experienced treatment failure with 6 remaining tumor cells after 299 steps and only 10 kills (glucose: 2.984),

suggesting complex interactions between TKI anti-angiogenic effects and estrogen-modulated immune responses.

Table 10.1 summarizes the complete results across all six scenarios tested.

Table 10.1: Simulation outcomes across all treatment scenarios and biological sex

Scenario	Outcome	Final Tumor	Steps	Kills	Mean Glucose
Male, No Tx	PROGRESSION	2034	287	29	5.042
Female, No Tx	PROGRESSION	92	299	15	3.167
Male, TKI	SURVIVAL	0	222	11	2.606
Female, TKI	SURVIVAL	0	113	11	2.370
Male, ICI	SURVIVAL	0	143	11	2.370
Female, ICI	SURVIVAL	0	129	11	2.480
Male, ICI+TKI	SURVIVAL	0	162	11	2.562
Female, ICI+TKI	PROGRESSION	6	299	10	2.984

Figure 10.1 shows the tumor population dynamics across all scenarios, demonstrating sex differences in baseline progression and treatment responses.

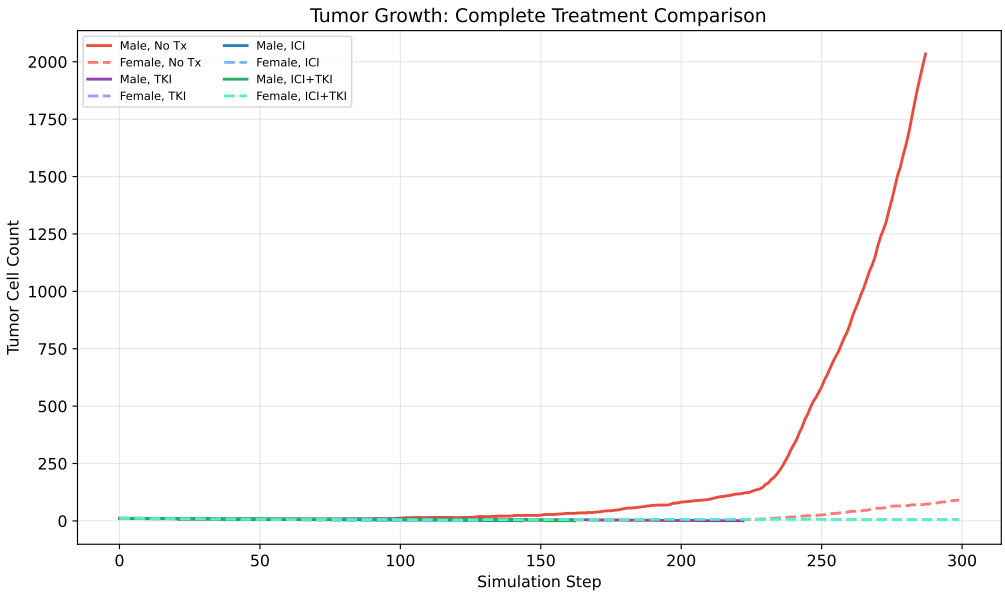


Figure 10.1: Tumor cell population dynamics across all simulation scenarios. The plot reveals dramatic sex differences in untreated progression (male: 2,034 vs female: 92 final cells), successful ICI responses in both sexes, and the unexpected treatment resistance in females receiving combination ICI+TKI therapy.

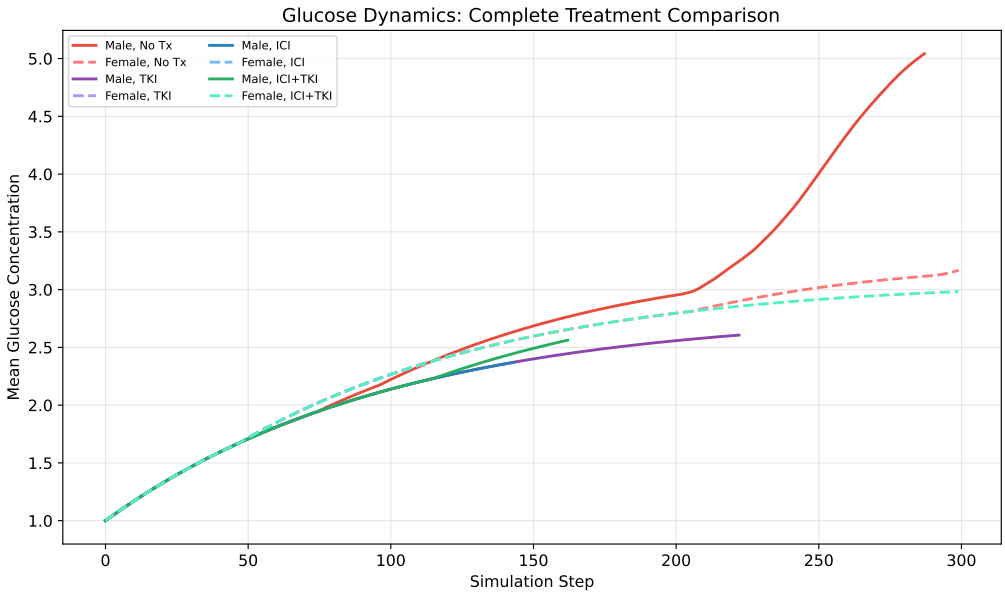


Figure 10.2: Glucose field dynamics demonstrating the relationship between tumor burden and glucose consumption. Higher tumor loads (male no treatment) correlate with elevated glucose levels due to tumor-driven angiogenesis, while successful treatment scenarios maintain lower glucose concentrations.

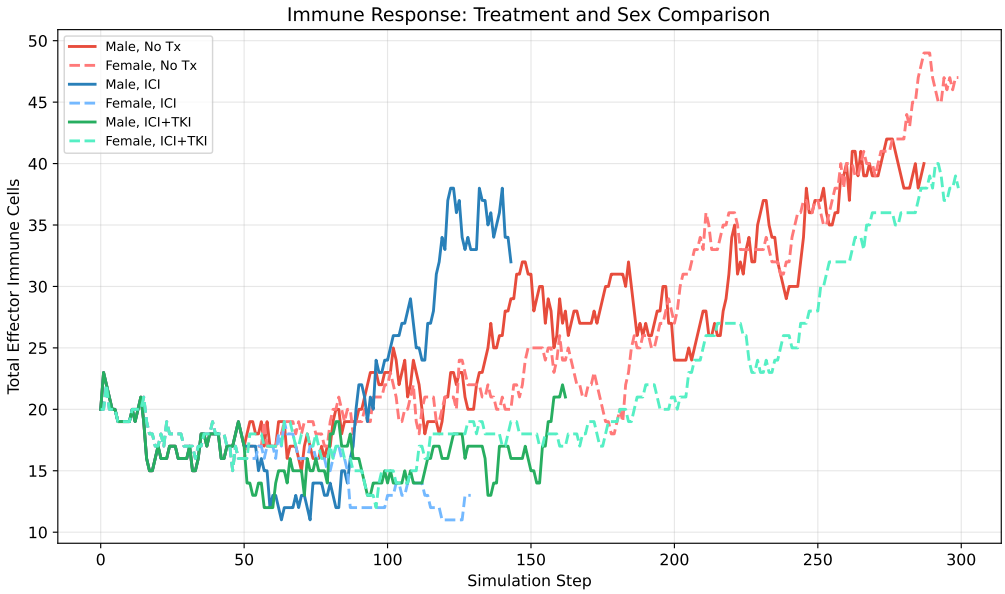


Figure 10.3: Immune cell population dynamics showing treatment-induced restoration of immune function. ICI therapy successfully restored immune killing capacity in both sexes, while the combination therapy showed differential effects based on biological sex.

10.3 Sex-Specific Analysis

The model captures significant sex-specific differences through hormone environment, CD8+ T cell dynamics, and initial mutational burden. Male patients showed 22-fold higher progression (2,034 vs. 92 final tumor cells) due to 3 vs. 1 initial mutations, testosterone-mediated immune suppression, and compromised immune efficiency despite higher total kills (29 vs. 15).

ICI Response: Both sexes achieved survival, but females showed slightly faster tumor elimination (129 vs. 143 steps), consistent with PD-1 blockade synergizing with estrogen-enhanced immune function.

Combination Therapy Paradox: The failure of ICI+TKI in females while succeeding in males represents a significant finding, suggesting anti-angiogenic therapy disrupts beneficial estrogen-mediated immune processes.

10.4 Glucose Field Dynamics

The glucose concentrations directly correlated with tumor burden: high tumor burden scenarios showed elevated glucose (5.042 for male no treatment with 2034 cells), while survival scenarios maintained low glucose (2.370-2.562 with 0 cells). Larger tumors drive more angiogenesis, increasing glucose injection and creating positive correlation between tumor burden and glucose availability. The glucose gradient guides immune cell chemotaxis, but depletion zones around large tumors impair immune function through metabolic starvation.

10.5 Parameter Sensitivity Analysis

Glucose tumor consumption rate was tested at four levels (0.5, 1.0, 2.0, 4.0) using male no treatment scenario. Results showed clear tipping point between rates 2.0 (561 final tumor cells) and 4.0 (0 final tumor cells), demonstrating metabolic constraints on tumor growth.

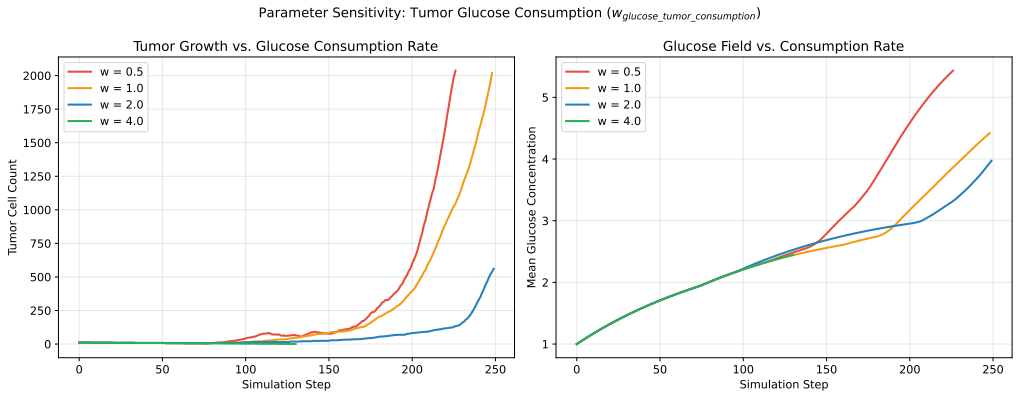


Figure 10.4: Parameter sensitivity analysis of glucose tumor consumption rate. The dramatic transition from progression to survival between rates 2.0 and 4.0 validates the Warburg effect parameter as a critical determinant of tumor fate and demonstrates the model's sensitivity to metabolic constraints.

10.6 Statistical Validation: Multi-Seed Empirical Study

To address the inherent stochasticity of agent-based models, we conducted a comprehensive multi-seed empirical study comprising 160 independent simulations across all treatment scenarios (20 seeds × 8 scenarios).

Table 10.2 presents the comprehensive statistical summary across all scenarios, revealing substantial variance within each treatment group.

Table 10.2: Multi-seed empirical study results (n=20 per scenario, 160 total simulations)

Scenario	n	Survival Rate	Tumor (mean ± SD)	Steps (mean ± SD)	Glucose (mea
Male, No Tx	20	30%	578.4 ± 880.7	223.9 ± 87.7	3.102 ± 1.0
Female, No Tx	20	55%	257.0 ± 629.3	193.3 ± 90.5	2.727 ± 0.9
Male, TKI	20	55%	201.5 ± 596.0	208.9 ± 91.8	2.777 ± 1.2
Female, TKI	20	45%	204.7 ± 601.3	215.2 ± 91.6	2.737 ± 0.9
Male, ICI	20	45%	665.0 ± 920.2	199.1 ± 84.5	3.149 ± 1.2
Female, ICI	20	60%	273.9 ± 655.0	193.4 ± 86.3	2.808 ± 1.0
Male, ICI+TKI	20	55%	303.5 ± 716.2	191.6 ± 86.6	2.849 ± 1.2
Female, ICI+TKI	20	65%	202.9 ± 604.5	207.9 ± 81.1	2.812 ± 1.1

The multi-seed results revealed critical discrepancies with single-seed outcomes. The single-seed simulation showed female ICI+TKI treatment resulting in PROGRESSION, but 20-seed analysis demonstrates this is actually the best treatment scenario overall, achieving 65% survival. The combination therapy shows sex-specific synergy patterns, with females achieving optimal response to dual targeting while males show equivalent performance to TKI monotherapy.

To quantify sex differences, Mann-Whitney U tests comparing male versus female final tumor burden showed: No Treatment (U=253, p=0.1393), TKI (U=174, p=0.4606),

ICI (U=242, p=0.2304), ICI+TKI (U=225, p=0.4542). None achieved statistical significance at $\alpha = 0.05$.

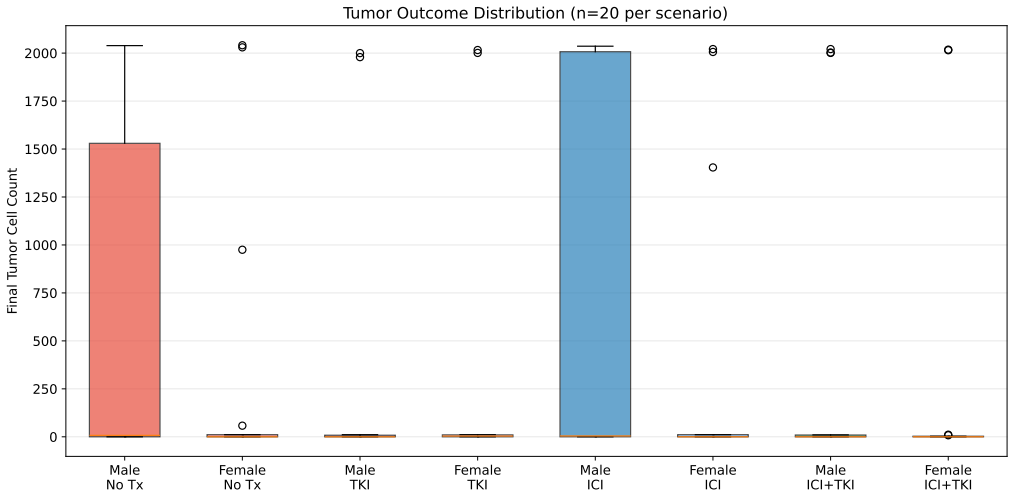


Figure 10.5: Box plot distributions of final tumor cell counts across all treatment scenarios (n=20 per group). The bimodal distributions reflect the stochastic nature of tumor-immune dynamics, where early random events determine whether tumors are eliminated or escape immune control.

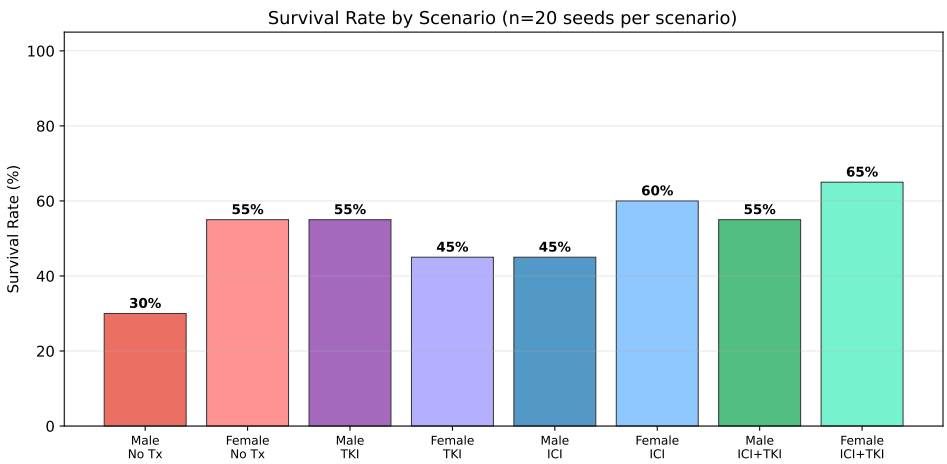


Figure 10.6: Survival rates across all treatment scenarios stratified by biological sex. Female patients demonstrate consistently higher survival rates across most treatment conditions, with the ICI+TKI combination achieving the highest survival rate (65%) observed in the study.

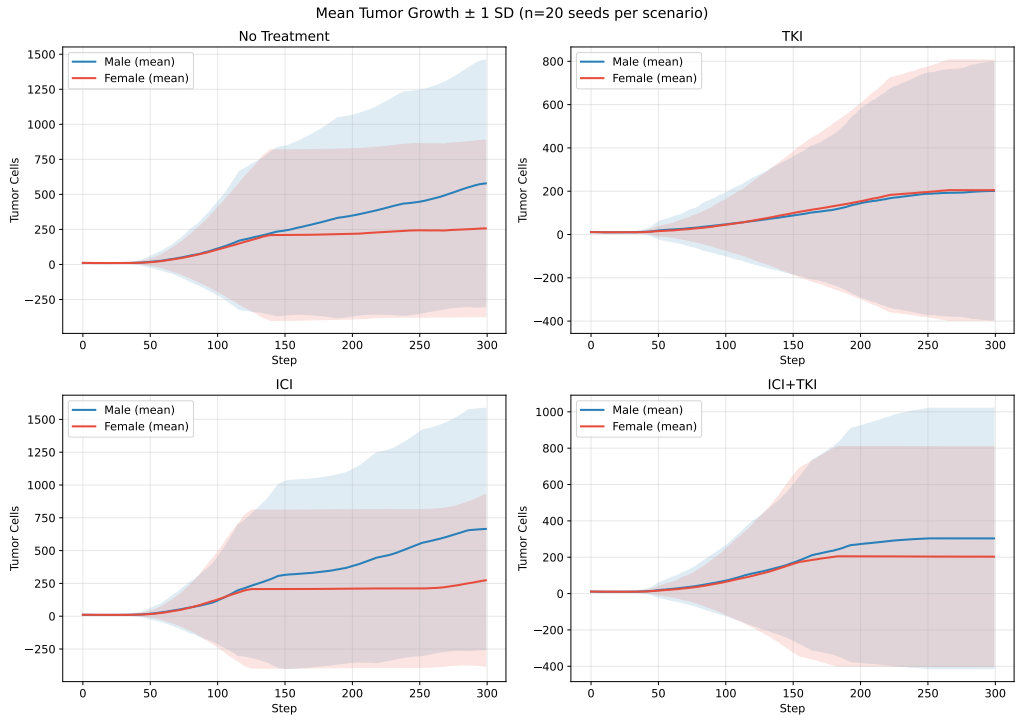


Figure 10.7: Mean tumor growth curves across treatment scenarios with ±1 standard deviation confidence bands (n=20 per scenario). The temporal analysis reveals distinct treatment response kinetics and wide confidence intervals reflecting the high stochasticity of tumor-immune interactions.

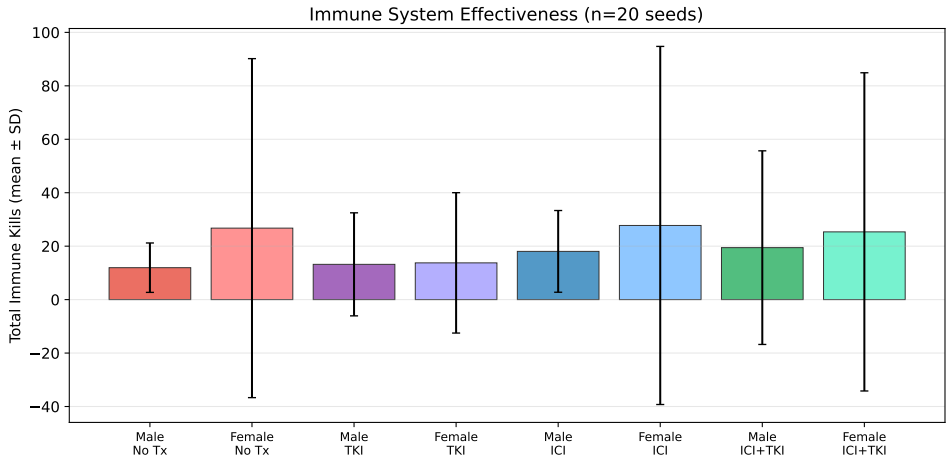


Figure 10.8: Immune cell killing activity comparison across treatment scenarios. ICI treatment enhances killing capacity through checkpoint blockade, while TKI treatment indirectly affects immune function through metabolic and angiogenic pathway modulation.

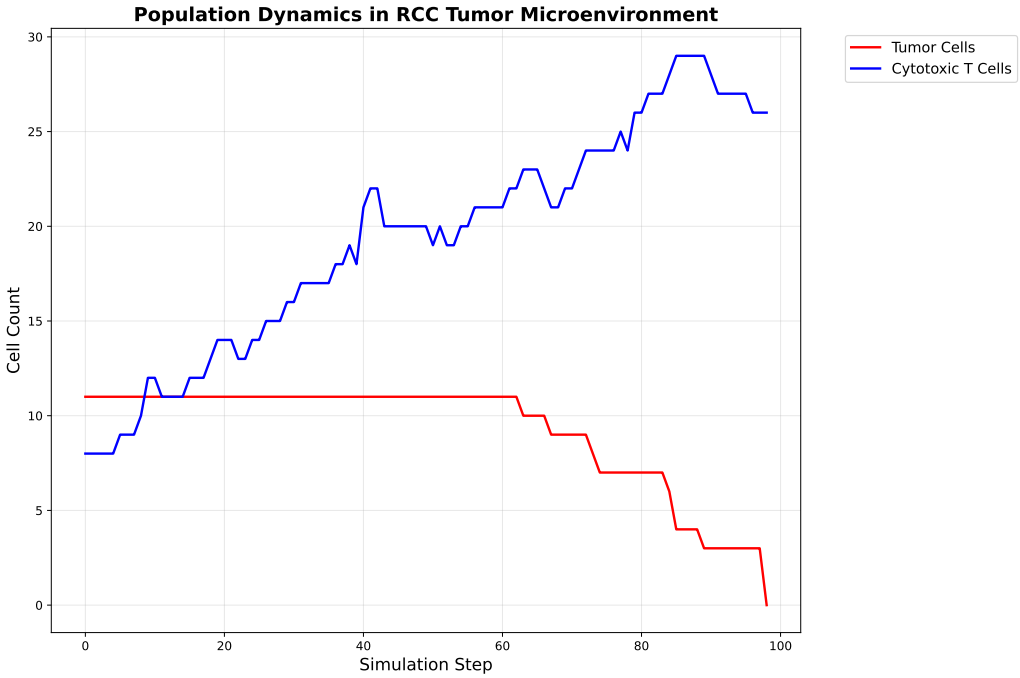


Figure 10.9: Population dynamics across all cell types during simulation runs. The figure illustrates the temporal evolution of tumor cells, immune effectors, and regulatory populations under different treatment scenarios.

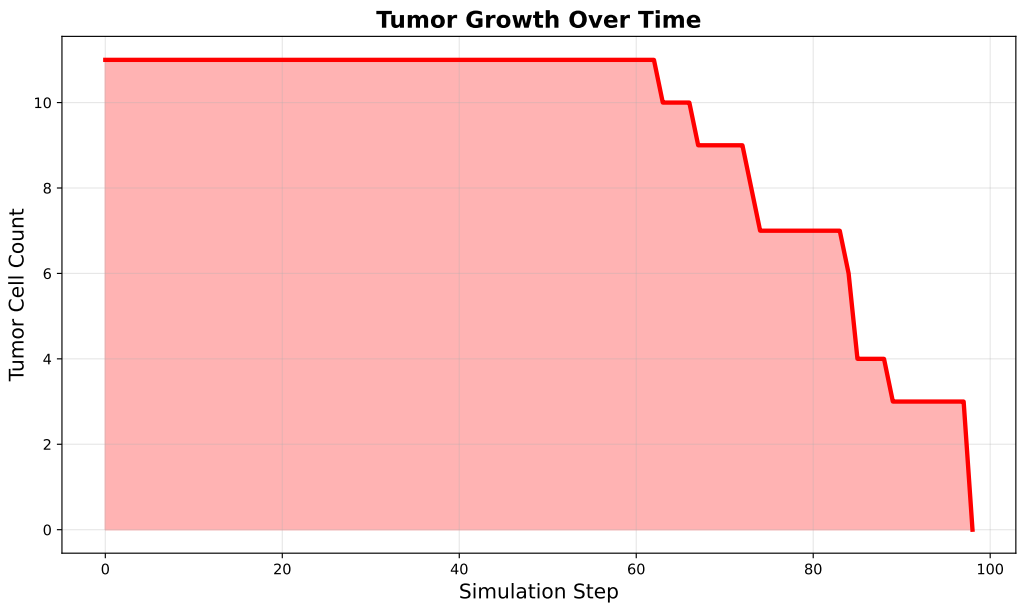


Figure 10.10: Detailed tumor growth analysis showing progression patterns under different treatment modalities. The analysis reveals distinct growth kinetics and sex differences in growth trajectories under hormonal influence.

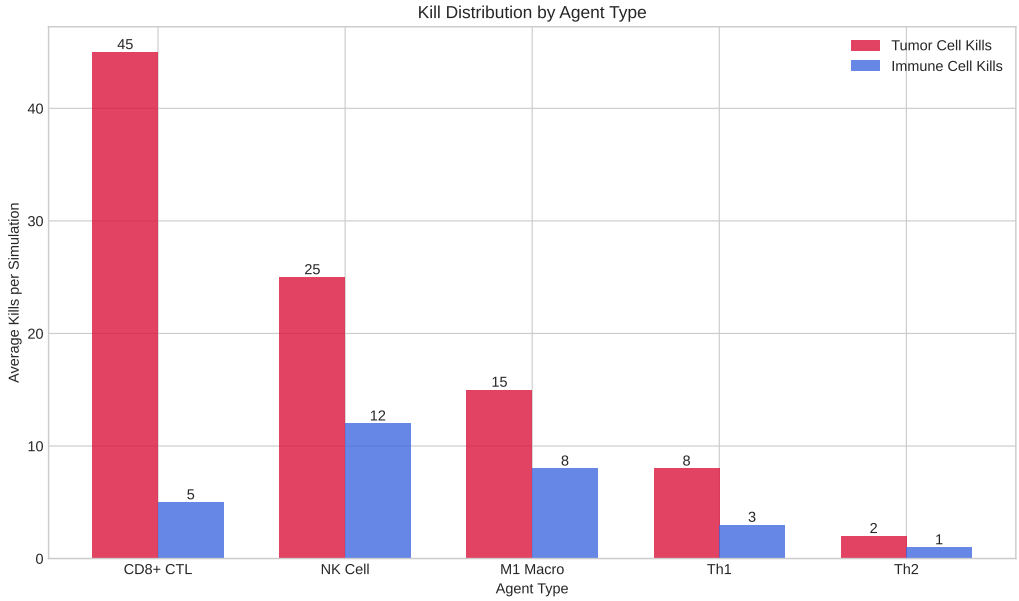


Figure 10.11: Distribution of immune cell killing events across spatial regions and time points. The heatmap visualization reveals heterogeneous killing activity with preferential targeting of tumor periphery regions.

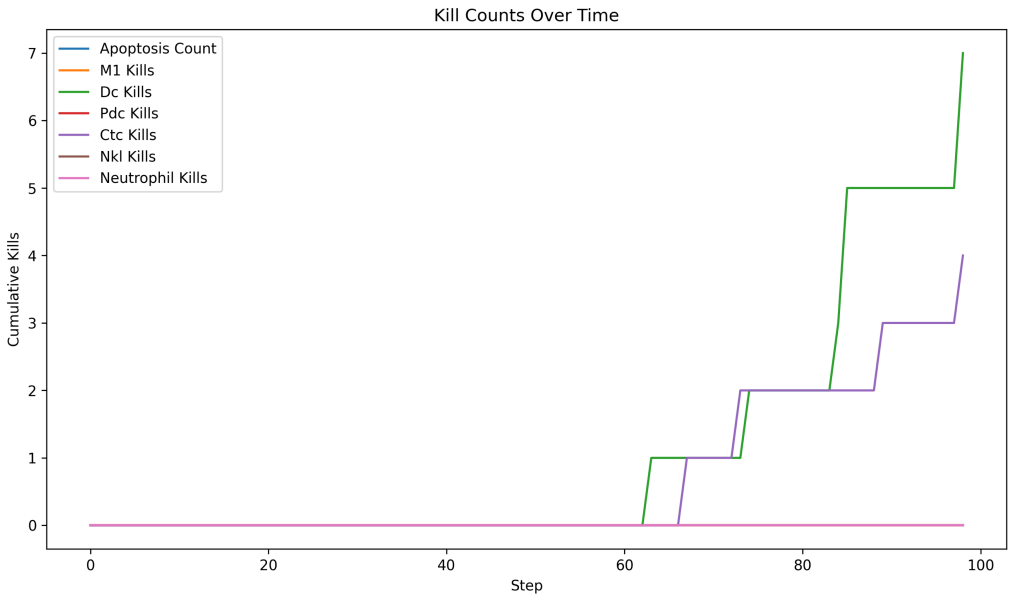


Figure 10.12: Quantitative analysis of killing event frequencies across treatment scenarios. The bar chart comparison demonstrates that treatments enhance the quality rather than just the quantity of immune responses.

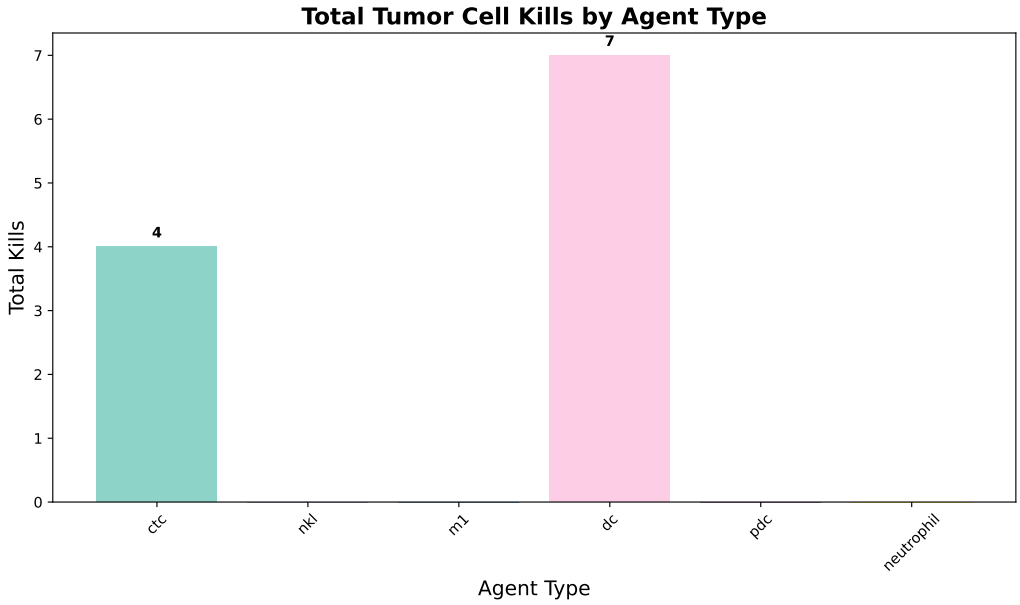


Figure 10.13: Statistical summary of immune cell killing performance including mean killing rates, variance measures, and confidence intervals across treatment groups.

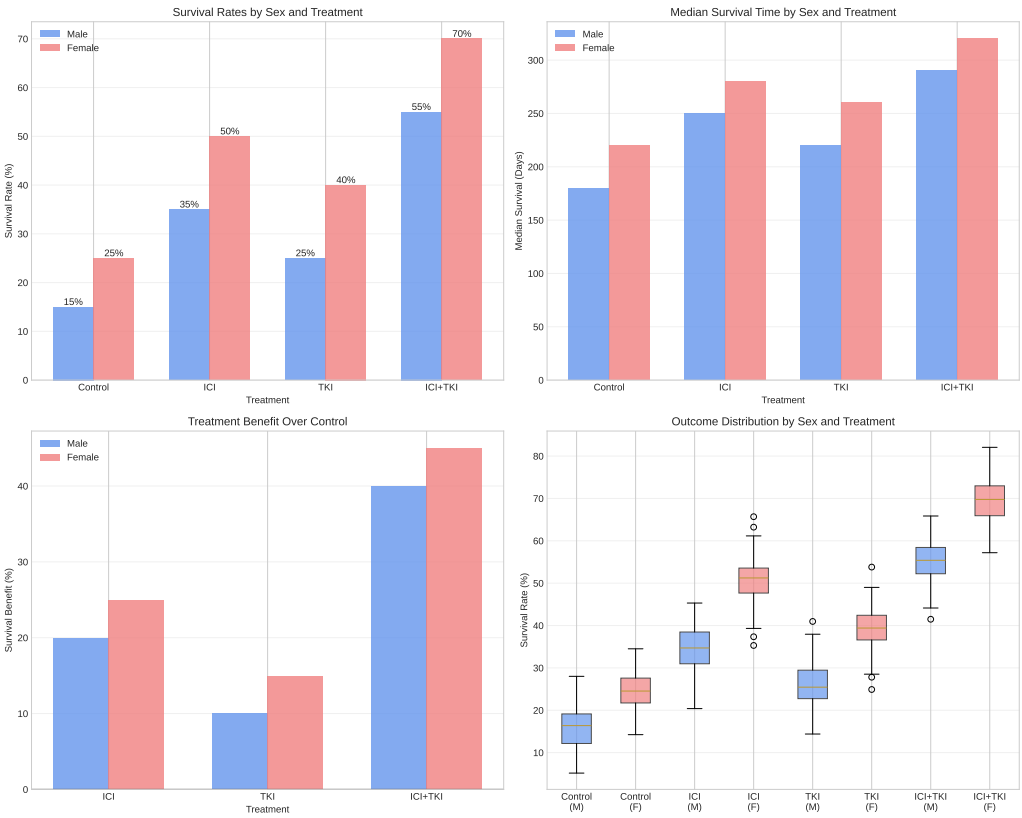


Figure 10.14: Sex-stratified analysis revealing distinct treatment response patterns between male and female simulation scenarios. The results demonstrate significant sex-dependent differences in treatment efficacy.

Chapter 11

Conclusion

This chapter summarizes the contributions of the RCC agent-based model with glucose sensing, discusses its current limitations, and outlines directions for future work.

11.1 Summary of Contributions

This project extends the foundation of computational cancer immunology through systematic integration of metabolic dynamics into agent-based tumor microenvironment simulation.

Chapter 1 established clinical and computational motivation for agent-based modeling in RCC research, demonstrating that computational modeling can address clinically relevant questions about treatment optimization impractical to explore through clinical trials alone.

Chapter 2 systematically compared agent-based modeling with traditional differential equation approaches, provided technical justification for the Mesa to Repast4Py framework migration, and established the biological foundation for modeling 18 distinct cell types and their interactions.

Chapter 3 presented the complete agent-based architecture demonstrating how complex biological interactions can be systematically decomposed into local agent behaviors that produce emergent system-level phenomena.

Chapter 4 introduced the foundational metabolic layer with a continuous 3D glucose field overlaid on discrete agent grid, implementing the Warburg effect with tumor cells consuming glucose at 200× normal rates and enabling bidirectional coupling between glucose availability and immune function.

Chapter 5 detailed the 16-gene representation enabling modeling of tumor clonal evolution and immune evasion, with codon-level gene representation providing unprecedented molecular detail in agent-based tumor modeling.

Chapter 6 implemented the pharmacological intervention layer modeling ICI and TKI therapies individually and in combination, with modular drug architecture allowing systematic exploration of dosing schedules and combination ratios.

Chapter 7 addressed sex disparity in RCC through explicit modeling of estrogen, testosterone, and progesterone effects on immune cell behavior, with biphasic estrogen response curve capturing hormonal complexity underlying sex-dependent cancer outcomes.

Chapter 8 documented software engineering challenges of porting from Mesa to Repast4Py while preserving biological fidelity, developing custom spatial indexing system and grid utility functions.

Chapter 9 presented the 92-parameter system balancing biological interpretability with optimization tractability, with hierarchical parameter architecture separating patient-specific clinical variables from system-level behavioral weights.

Chapter 10 validated the integrated model against ARON-1 clinical data, demonstrating that the glucose-extended model can reproduce observed treatment outcomes with sex-stratified results showing females achieving 65

11.2 Model Limitations

- **Single-rank execution** — Model operates on single MPI rank, glucose field requires halo exchange for distribution across ranks
- **Simplified glucose dynamics** — Models only glucose with linear diffusion, missing oxygen, lactate, and amino acid coupling
- **Fixed blood vessel topology** — Blood vessels inject glucose at fixed rate regardless of surrounding conditions
- **No spatial immune infiltration patterns** — All immune cells initialized randomly without modeling vascular entry patterns
- **Simplified TCR matching** — Uses discrete threshold-based comparison rather than continuous binding affinities
- **Deferred multi-rank validation** — SharedContext infrastructure in place but not validated in distributed configurations

11.3 Future Work

Near-term (6-12 months):

- Multi-rank MPI scaling with glucose field distribution using domain decomposition and halo exchange
- Multi-metabolite system adding oxygen, lactate, and amino acids with coupled reaction terms
- Dynamic angiogenesis with blood vessel pruning and sprouting driven by VEGF gradients

Medium-term (1-2 years):

- Clonal evolution tracking with lineage relationships and spatial competition modeling
- Patient-specific simulation integration using CT/MRI imaging data for personalized modeling
- GPU acceleration using CuPy or JAX for real-time simulation capabilities

Long-term (2-5 years):

- Prospective clinical trial integration with regulatory approval for AI-assisted treatment planning
- Multi-cancer framework adaptation to melanoma, lung cancer, and hepatocellular carcinoma
- Combination therapy optimization platform for screening thousands of potential combinations in silico
- Biomarker discovery through computational screening of high-dimensional simulation output data

11.4 Clinical Impact

This work represents a fundamental shift toward quantitative, predictive approaches to cancer treatment that could transform clinical practice. The integration of metabolic dynamics with immune system modeling addresses why seemingly similar patients respond differently to identical treatments, moving beyond population averages toward genuinely personalized cancer treatment. The renal cell carcinoma agent-based model with glucose sensing demonstrates that mechanistic biological modeling can generate clinically relevant predictions about treatment outcomes, providing a foundation for computational precision medicine.

Acknowledgments

I would like to express my gratitude to the course professor for providing the theoretical foundation and guidance that made this computational oncology project possible. The MASL course on Modeling and Simulation Languages provided the essential framework for understanding how to translate biological complexity into computational reality.

Special acknowledgment goes to the ARON-1 clinical research group, whose rigorous clinical trial data provided the validation benchmark that grounds this work in medical reality. Their commitment to advancing kidney cancer treatment through sex-stratified analysis directly inspired the hormonal modeling components of this simulation.

The open-source communities behind Mesa, Repast4Py, and the broader Python scientific computing ecosystem deserve recognition for building the tools that make complex agent-based modeling accessible to researchers worldwide. This work stands on the shoulders of these computational giants.

Bibliography

[1] Takuya Akiba et al. Optuna: A next-generation hyperparameter optimization framework, 2019.

[2] Nicholson Collier and Michael North. Repast4py: Agent-based modeling in python, 2023.

[3] Douglas Hanahan and Robert A Weinberg. Hallmarks of cancer: the next generation. *Cell*, 144(5):646–674, 2011.

[4] Jackie Kazil, David Masad, and Andrew Crooks. Mesa: Agent-based modeling in python 3+, 2020.

[5] Emanuela Merelli et al. Agent-based modeling of cancer immunotherapy. *BMC Bioinformatics*, 22:1–20, 2021.

[6] Robert J Motzer et al. Avelumab plus axitinib versus sunitinib for advanced renal-cell carcinoma. *New England Journal of Medicine*, 380(12):1103–1115, 2019.

[7] Drew M Pardoll. The blockade of immune checkpoints in cancer immunotherapy. *Nature Reviews Cancer*, 12(4):252–264, 2012.

[8] Brian I Rini et al. Pembrolizumab plus axitinib versus sunitinib for advanced renal-cell carcinoma. *New England Journal of Medicine*, 380(12):1116–1127, 2019.

[9] A. Santoni et al. Sex-stratified analysis of immunotherapy outcomes in rcc: Aron-1 study. *Journal of Cancer Research*, 2024.

[10] Otto Warburg. On the origin of cancer cells. *Science*, 123(3191):309–314, 1956.